

1. INTRODUCTION

AI is everywhere now, changing how we use tech. Automated tools are a must-have in fields like content creation and data analysis.

1.1 PROJECT DESCRIPTION

Quick AI is a neat platform that gathers a bunch of smart tools in one place. It's made to help you avoid boring stuff, work faster, and create cool content easily.

It uses machine learning for things like removing backgrounds, making images, getting rid of objects, checking resumes, writing articles, and creating blog titles.

The aim is to keep AI simple, even if you're not a techy. Quick AI can grow with you and is easy to learn, making it good for personal or work projects, whether you're a designer, writer, or looking for a job.

1.1.1 OBJECTIVES

Here's the deal with Quick AI:

1. We're using AI to automatically do stuff like edit photos and write things.
2. It's made to be super simple to use, even if you're not tech-savvy.
3. It's designed to help you get more done with content, resumes, and pictures.
4. It's built strong, so it can handle lots of people using it at the same time.
5. It puts a bunch of AI tools in one place, which is handy.

Basically, Quick AI is here to save you time and help you get great stuff done.

1.1.2 SYSTEM OVERVIEW

Okay, so Quick AI is basically an AI platform made of different pieces. Here's the breakdown:

- **Presentation Layer:**
 - The website people use in their browsers.
 - Simple to use with easy controls.

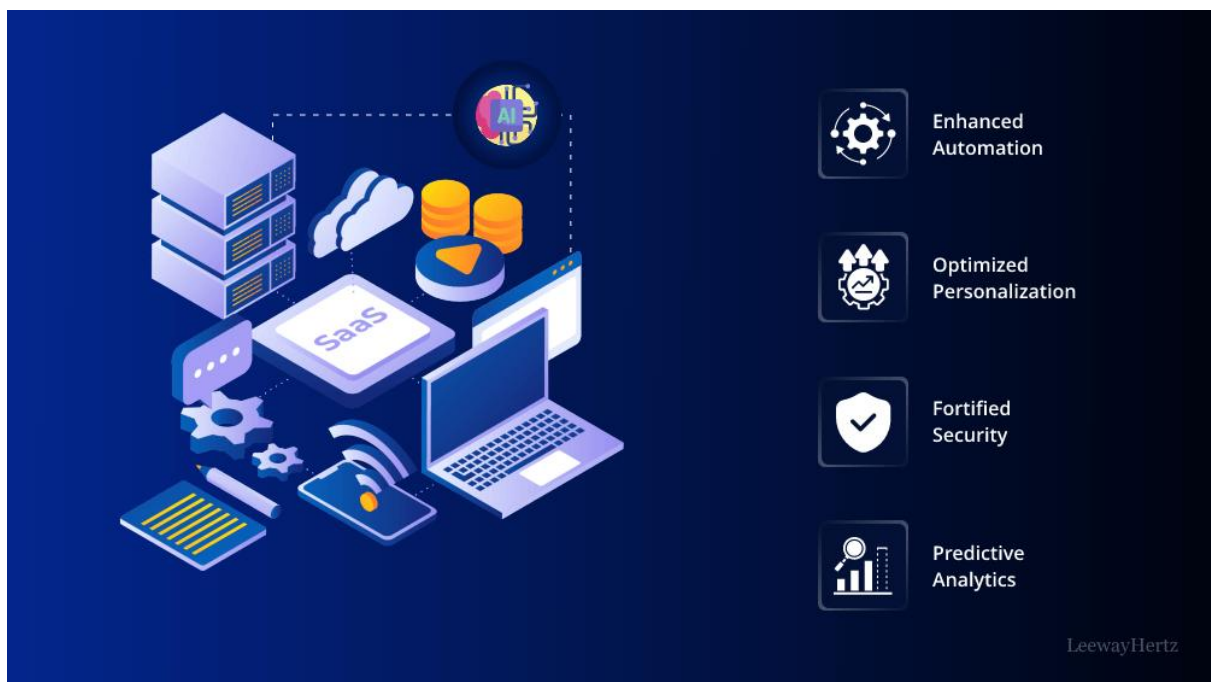
- **Application Layer:**

- The behind-the-scenes stuff that handles requests, runs the AI, and keeps things flowing.
- Uses APIs to talk between the website and the backend

- **Data and AI Layer:**

- Storage for user data and what the AI makes.
- The AI models that do things like image processing and writing.

This setup makes the system stable, able to grow, and easy to keep running, so we can add more AI stuff later.



1.1.3 Modules and Functionalities

Background Remover

- Automatically takes backgrounds out of photos using fancy AI.
- Works with most image types.

Image Generator

- Makes images based on what you type in.
- Can make realistic or artistic stuff.

Object Remover

- Removes anything you do not want in a picture and makes it look natural.

Resume Analyzer

- Checks resumes for mistakes and suggests ways to improve them.
- Gives tips to help you get through those automated application systems.

Article Generator

- Writes good articles on any topic.
- Uses AI to make sure the articles make sense.

Blog Title Generator

- Comes up with catchy blog titles.
- Helps people find your content online.

1.1.4 Technology Stack

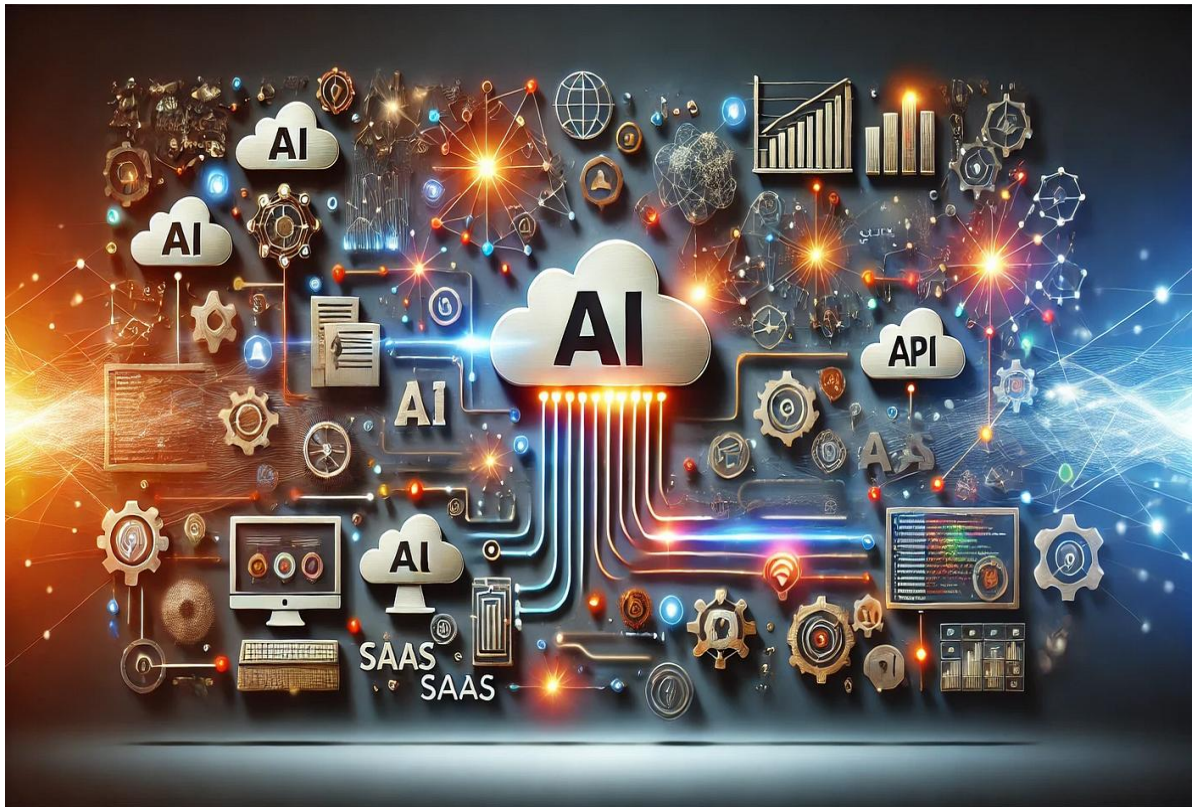
- Quick AI uses modern tech to be reliable:
- Frontend: HTML, CSS, JavaScript, and React.js for a nice, interactive website.
- Backend: Node.js with Express for handling requests and running the platform.
- Database: PostgreSQL for storing data.
- Cloud & Hosting: AWS/Azure for storage and scaling.

This mix of tech makes sure everything runs smoothly and securely.

1.1.5 Methodology

We're using Agile to build Quick AI, which means we're working step by step and getting feedback along the way. Here's the process:

- 1. Requirement Analysis:** Figuring out what the platform needs to do.
- 2. System Design:** Planning out how everything will fit together.
- 3. Implementation:** Writing the code for the website, backend, and AI parts.
- 4. Testing:** Making sure everything works right.
- 5. Deployment:** Putting the platform on the cloud.
- 6. Maintenance and Updates:** Keeping things running and adding new stuff based on feedback. This way, we get a good product that people like.



1.1.6 Expected Outcomes

Here's what we expect to achieve with Quick AI:

- Simple content creation and image editing. More work done by individuals and businesses.
- Easy access to AI tools, even if you are not techy. Able to handle more users as we grow.
- One platform with a bunch of AI tools inside, so you do not need different apps.
- This all goes back to our goal of making a complete and smart platform.

1.2 COMPANY PROFILE

- Name: Mobinius Technologies Private Limited
- Started: September 26, 2011
- Type: A regular company

- Where: A-Block, 3rd Floor, Brigade Software Park, No. 42, 27th Cross, Banashankari 2nd Stage, next to Pai Vista Convention Hall, Bengaluru, KA 560070.

1.2.1 Leadership & Corporate Details:

- Founders/Directors: Damodar Puthiya and Lokesh Mullamajalu.
- Money Stuff: They had ₹5.0 lakhs authorized capital; paid-up capital was ₹1.0 lakh.
- Status: Still going strong.

1.2.2 What They Do — Services & Expertise

Mobinius helps other companies with tech stuff like:

- Making phone apps (Android, iOS)
- Building websites
- Designing interfaces
- IoT (connecting devices)
- AI, data analysis
- Cloud stuff
- Testing, product development

They work with all kinds of clients, from small businesses to big companies. They have worked in healthcare, retail, logistics, education, and more.

1.2.3 Size & Market Position

- Employees: Somewhere between 100 and 500 people.
- Rates: Around \$25-49/hour for some services.
- Recognition: Known in the blockchain and IoT fields.

1.2.4 Strengths & Challenges (Based on Reviews / Feedback)

Strengths:

- Can do a lot of different tech things, from design to deployment.
- Works in many industries.
- Known for IoT and blockchain.

Challenges:

- Some reviews say they sometimes have trouble finding the right people for the job.
- Communication glitches flagged in reviews.

2. LITERATURE SURVEY

AI and SaaS are changing how businesses work. Lots of AI tools exist, like Canva for backgrounds, Grammarly for writing, and ChatGPT for content. But most of these are separate apps, so people have to use a bunch of different things to get stuff done. People want everything in one place while keeping quality high. SaaS platforms are good for AI because they can grow easily and do not cost too much. Past research also shows problems like keeping data safe and making sure AI is correct. So, there is a need for a single AI platform that can do lots of things for different users.

2.1 Existing and Proposed System

Existing System:

AI tools are good but separate. Each one needs its own account and has its own learning curve. This makes things complicated when you are switching between different tools to complete a project, which wastes time and money.

Proposed System:

Quick AI puts all the main AI into one platform. You can remove backgrounds, make images, get rid of objects, check resumes, write articles, and more. It is on the cloud, so it is available everywhere and can grow as needed. This saves time and gives you a smooth experience with AI.

2.2 Feasibility Study

We need to make sure Quick AI can actually be built and kept running.

2.2.1 Technical Feasibility: It can be done with the tech we are using (PostgreSQL, Express, React, Node.js). It is a good foundation for web development. This tech works well with AI and cloud platforms, so it is a good, safe, and doable solution.

2.2.2 Operational Feasibility: It is easy to use and well-organized, so people will use it. It also fills a gap in the market by putting all the tools in one spot. We can keep data safe. So, the project works in every way.

2.2.3 Economic Feasibility: Can we build and maintain it without spending too much? Since it is SaaS, we do not need to invest in tons of equipment. We can pay for cloud

services as we go. We can make money with subscriptions and services. So it can happen on a budget.

2.2.4 Legal Feasibility: Quick AI needs to follow the rules. Since we're processing user data, we need to follow data protection laws. We also need to make sure we're not stealing content with the AI. With strong data protection and good policies, Quick AI is doable.

2.3 Tools and Technologies Used

Quick AI uses new tech to be accurate and fast while being friendly to interact with.

Frontend:

- React.js – For a website that looks good and responds quickly.
- HTML5, CSS3, JavaScript – The core web stuff.

Backend:

- Node.js – For a platform that can handle a lot of users.
- Express.js – For managing requests.

Database:

- PostgreSQL – For storing user data.

Deployment and Cloud:

- AWS / Azure – For hosting and storage.
- Docker – To make it easy to put the platform on different servers.
- Git & GitHub – For working together on the code.

2.4 Hardware and Software Requirements

Hardware:

- Laptop with an Intel i5 or better.
- Qualcomm Snapdragon.
- At least 8-16 GB RAM.

Software:

- Windows 10/11
- Visual Studio Code.
- Node.js & npm – For the backend.
- PostgreSQL client tools – pgAdmin for managing the database.
- Browsers–Chrome, Edge, Firefox, Safari.

3. SOFTWARE REQUIREMENTS SPECIFICATION

This is how the Quick AI platform works. This document makes sure everyone knows what's expected and what it will be used for: building, putting it out there, and making it better in the future. Main goal? To build a platform that is solid, able to grow, and friendly to work with integrates AI tools for images and writing.

3.1 USERS

What's the point of the app?

- To build an AI platform on the cloud with a bunch of tools.
- To make things faster by automating tasks.
- To build a secure and scalable system that works well for everyone.

3.2 SCOPE AND OBJECTIVE

The app has six parts:

- Background Remover: Take out or change image backgrounds.
- Image Generator: Make images from text.
- Object Remover: Get rid of stuff you do not want in photos.
- Resume Analyzer: Check resumes and give tips.
- Article Generator: Write articles automatically.
- Blog Title Generator: Suggest blog titles.

It works in web browsers and allows multiple users. It can also grow easily.

3.3 PROBLEM STATEMENT

Quick AI has six main parts, each with its own AI tool. The Background Remover removes backgrounds from images. The Image Generator makes images from text. The Object Remover gets rid of things you do not want. The Resume Analyzer gives advice on resumes. The Article Generator writes articles. The Blog Title Generator comes up with blog titles. Each part works with the website to do its job. The design makes it easy to maintain and add new stuff later.

3.4 FUNCTIONAL REQUIREMENT

Quick AI needs to be able to do these tasks. Users need to be able to upload stuff for processing. Each part should use AI to process the uploads. The system needs to keep

accounts secure. Users need to be able to download the processed files. The system should keep track of what people are doing. The system also needs to handle errors and notifications.

3.5 NON-FUNCTIONAL REQUIREMENT

This part is about how the platform should work. It needs to be able to handle more users without slowing down. It needs to be available almost all the time. It needs to be fast. It needs to be secure. It needs to be easy to use. It needs to be easy for developers to make updates. It needs to follow the rules about handling data.

4. SYSTEM DESIGN

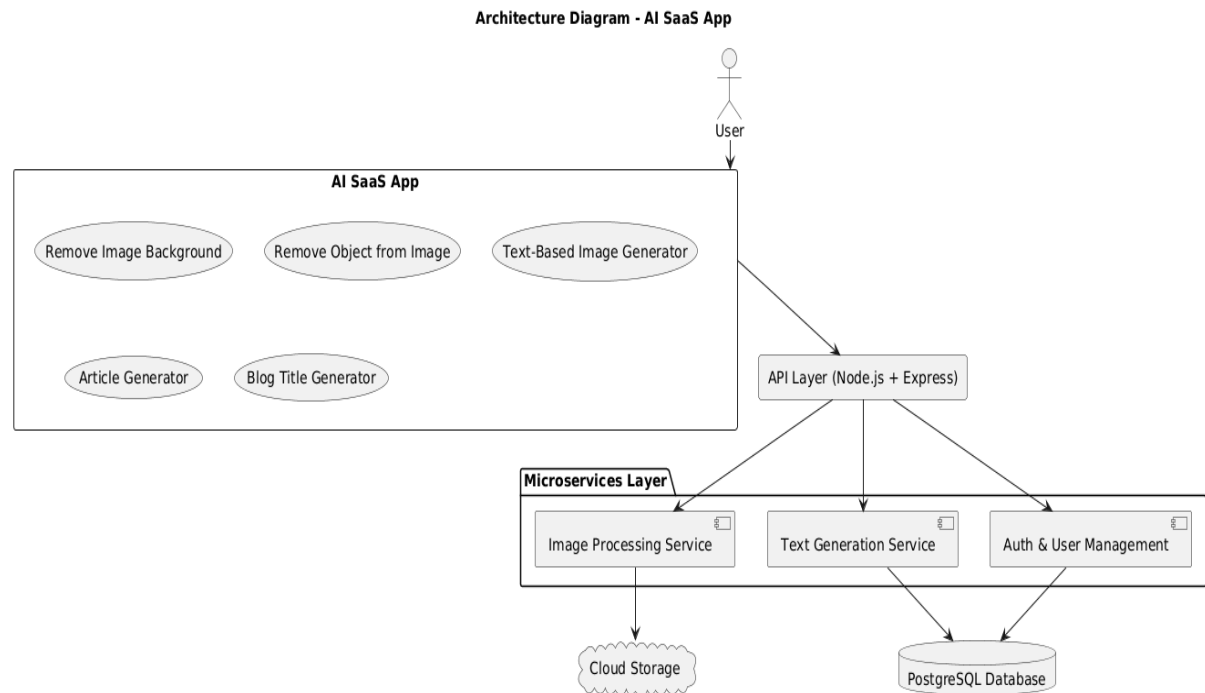


Figure 4.0

DESCRIPTION

AI SaaS App (Frontend - React)

What it does:

- The interface for interacting with all AI features.
- Upload images, enter text, and view the results.
- Handles accounts.

Techy Stuff:

- React + Context API
- Axios/Fetch for API calls
- Responsive design

API Gateway (Node.js + Express)

What it does:

- Connects the frontend and backend.
- Handles requests, validates, limits, and handles errors.
- Manages security

Techy Stuff:

- Express middleware
- JWT
- API versioning and modular route controllers.

Microservices Layer

Split into three to work better:

1. Image Processing Service**What it does:**

- Removes backgrounds and objects.
- Makes images from text.

Techy Stuff:

- Python (FastAPI or Flask) or Node.js with TensorFlow.js
- Models that are pre-trained.
- Handles tasks.

2. Text Generation Service**What it does:**

- Writes articles and blog titles.
- Handles formatting.

Techy Stuff:

- Node.js
- Integrates with OpenAI, Cohere, or others.
- Caches prompts to speed things up.

3. Auth & User Management Service**What it does:**

- Handles registration, login, roles, and permissions.
- Tracks usage and billing.

Techy Stuff:

- OAuth2, JWT, or Firebase Auth.
- Access control.
- Secure password stuff.

PostgreSQL Database**What it does:**

- Stores user data, image data, and billing.

Techy Stuff:

- Organized setup with indexes.
- Sequelize or Prisma ORM for Node.js.
- Backups and logging.

Cloud Storage (e.g., Cloudinary)**What it does:**

- Stores images

Techy Stuff:

- Pre-signed URLs
- Auto-deletion
- CDN

4.1 Architecture Diagram

Architecture Diagram - AI SaaS App

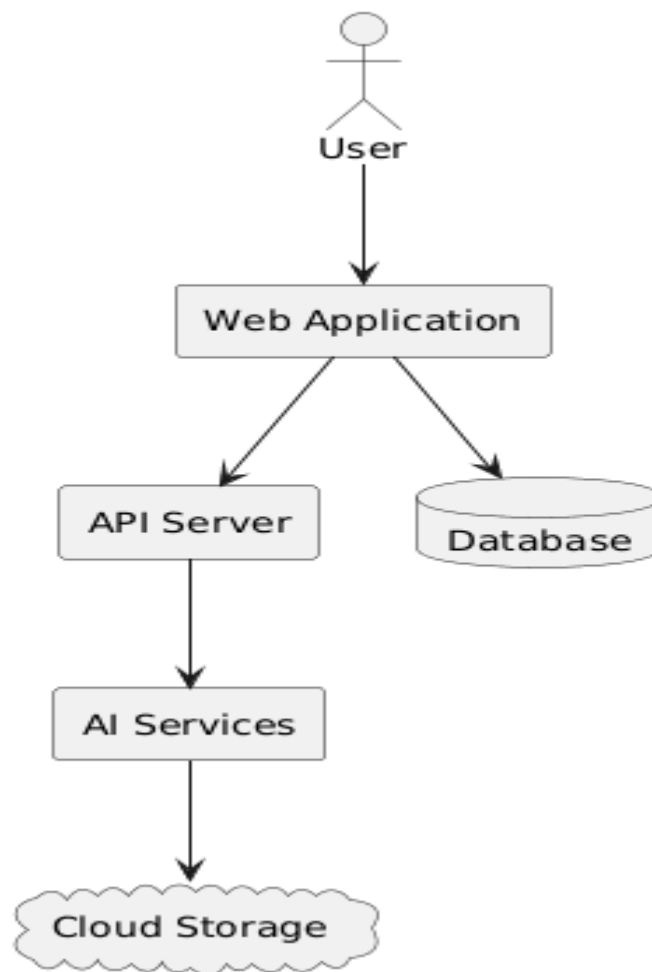


Figure 5.2

4.1.1 Purpose

- This diagram shows how our AI-powered SaaS application works. It starts with you, the user, using the web application. We usually build this with React.js so it looks good and is easy to use. You can upload pictures, type in what you want, and see what the application creates.
- When you send something from the web Application, it goes to the API server, which we built using Node.js and Express. Think of this server as the main controller. It checks who you are, makes sure your request is okay, and sends it to the right place. It talks to three main parts: the AI Services, where the AI magic happens; Cloud Storage, where we keep all the files; and the Database.

- The AI Services part handles machine-learning stuff like removing backgrounds from images, erasing objects, making images from text, and creating content. These services might be made with Node.js and work with outside tools or special models we have trained.
- Cloud storage (like Cloudinary) keeps uploaded pictures and things we make safe. The database (PostgreSQL) stores data in order, like user info, what they've made before, and how they use the service. This setup keeps things separate, makes it easy to grow, and helps us use data well. It is good for making AI content fast in a software service.

4.2 Context diagram

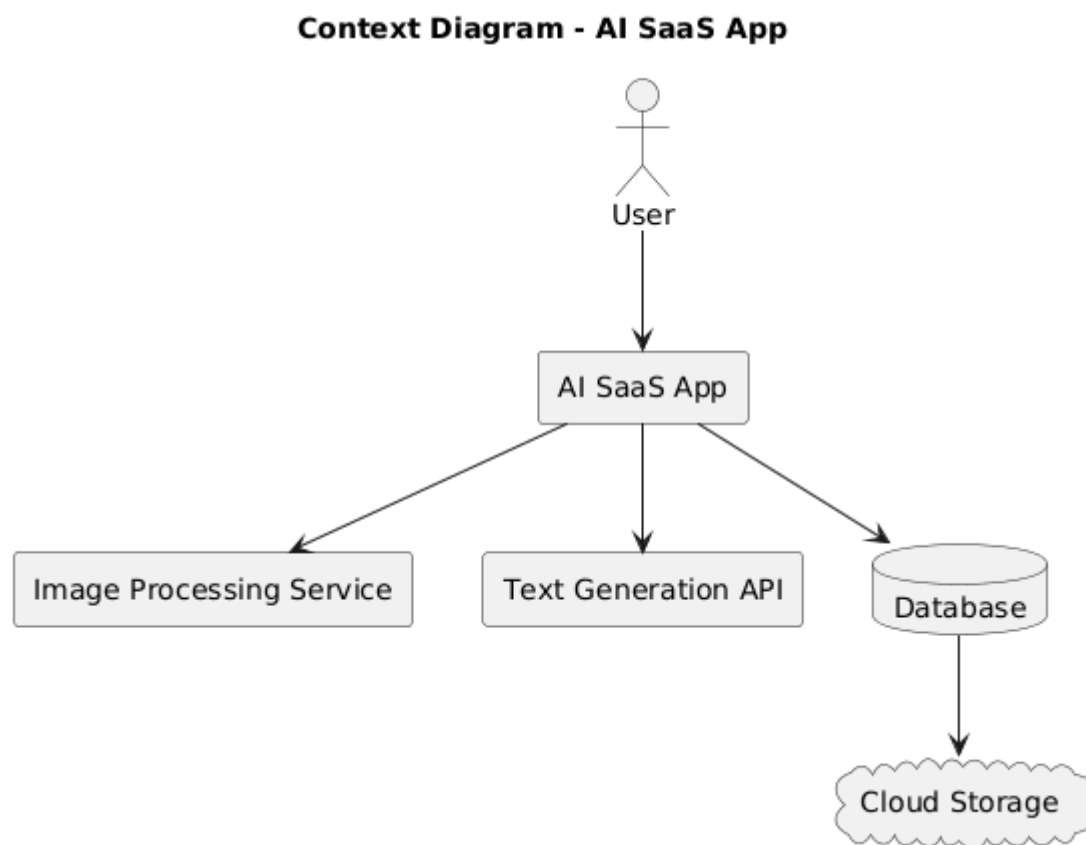


Figure 4.2

Purpose

This shows how the app works with other stuff.

Central System: AI SaaS App

This is where all the action is.

- Image background removal.
- Object removal from images.
- Generating images from text.
- Writing articles and blog titles.

1. User

- **Who:** The person using the app.
- **How:**
 - Uploads images.
 - Enters text.
 - Views content.
 - Manages their account.
- **What happens:** Sends requests and sees results.

2. Image Processing Service

- **What:** Handles image AI things.
- **Functions:**
 - Background removal.
 - Getting rid of objects.
 - Turning text into images.
- **What Happens:** Gets stuff from the app and sends back the finished stuff.

3. Text Generation API

- **What:** Generates text.
- **Functions:**
 - Writes articles and blog titles.
- **What Happens:** Gets instructions and sends back text.

4. Database (PostgreSQL)

- **What:** Stores data.
- **Contents:**
 - User info

- Image data
- Text logs
- Billing info
- **What Happens:** Supports analytics, dashboards, and user history.

5. Cloud Storage

- What: Stores images.
- Examples: Cloudinary.
- **What Happens:** uploads and provides URLs.

4.3 Data Flow Diagram

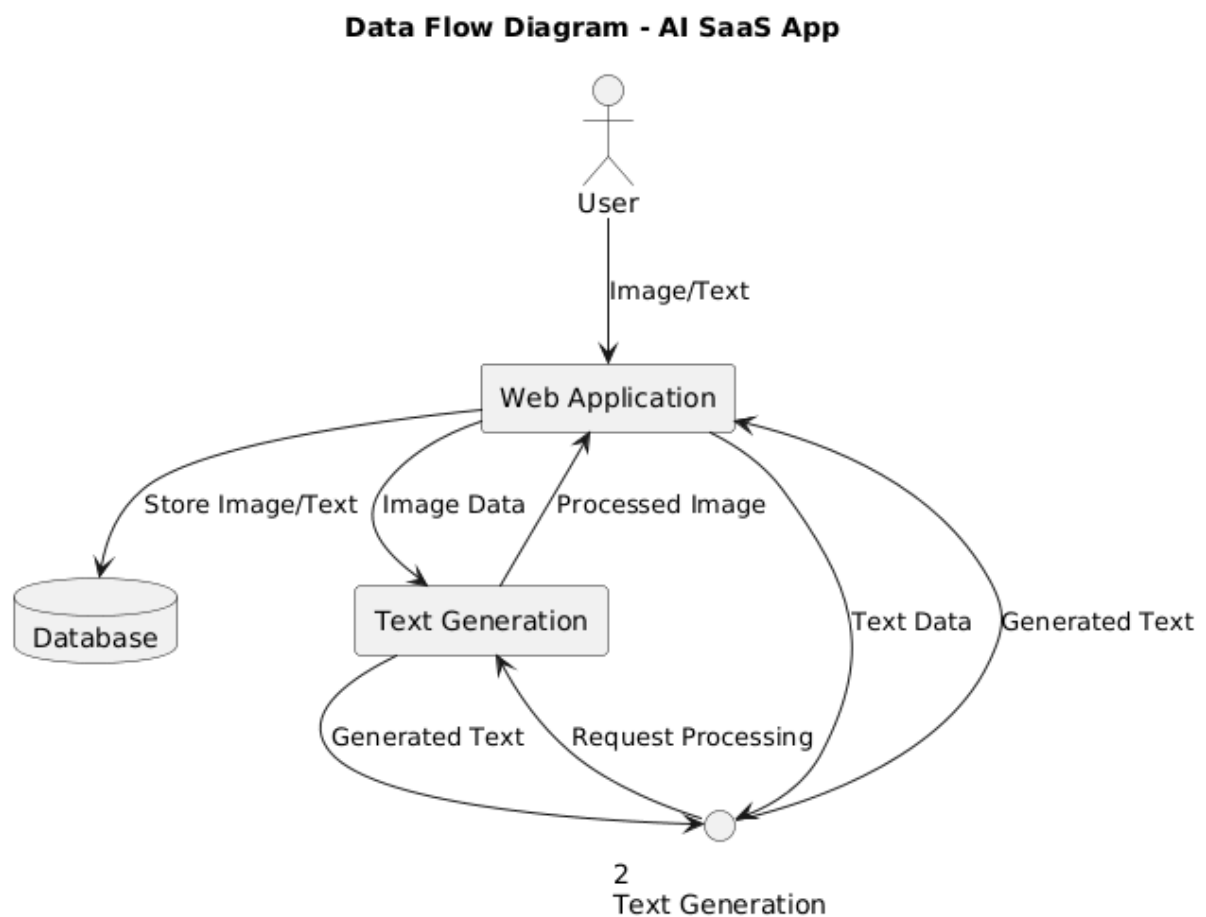


Figure 4.3

4.3.1 Purpose

Here's how data flows in our AI SaaS app: It shows the main actions, where data is kept, and outside pieces that handle user requests for making images and text.

5. DETAILED DESIGN

5.1 Class Diagram

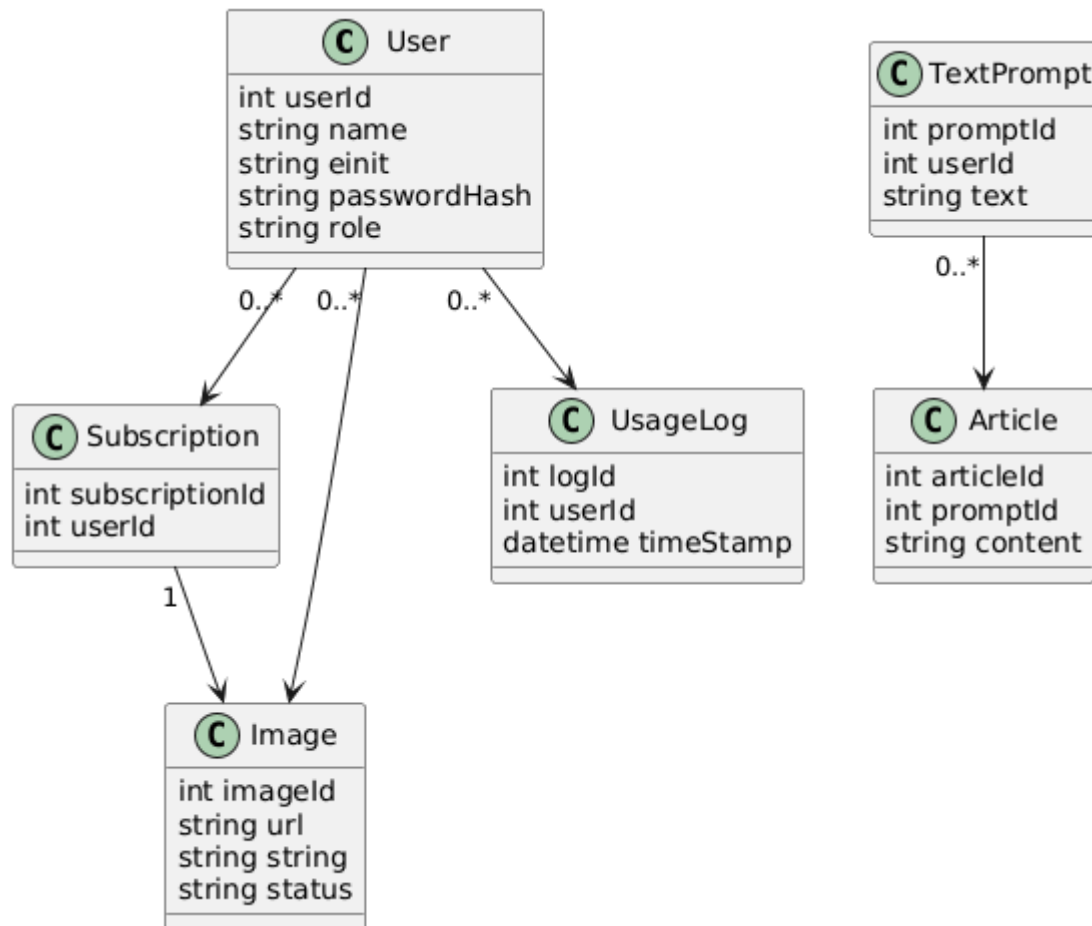


Figure 5.6

5.1.1 Purpose

The class diagram shows the object-oriented design of the AI SaaS application. It lays out the main parts of the system, what those parts do, and how they relate to each other. This helps the people coding the app to understand how data and actions are organized in different sections.

Here's a look at the main parts and what they do:

1. User

This represents someone who has signed up for the platform.

Info:

- User ID: A unique ID number.
- name: The user's full name.
- email: The user's contact email.
- Password Hash: A protected password.
- role: What kind of user they are (like admin, free, or premium)

Relations:

- One user can have many images, text prompts, and usage logs.
- One user has one subscription.

2. Image Data

This keeps track of images people upload or make.

Info it saves:

- Image ID: Each image gets its own ID.
- User ID: Who owns the image.
- URL: Where the image lives online.
- Status: Tells you if it's still processing or good to go.

3. Text Input

This saves the exact text people type in when they want to make something.

Info:

- Input ID: Each input gets a special ID.
- User ID: Tells you who wrote the input.
- Text: What the input actually says.

4. Article Data

This represents the articles made from the text inputs.

Info:

- Article ID: A unique ID for each article.
- Input ID: Which text input was used to create the article.
- Content: The actual text of the article.

5. Blog Title

This store's blog titles are made from prompts.

Info:

- Title ID: A unique title ID.
- Prompt ID: The prompt used.
- title: The blog title.

6. Subscription

This tracks each user's subscription plan and its expiration date.

It shows:

- Subscription ID: A unique ID for each subscription.
- User ID: The user linked to the subscription.
- Plan Type: The subscription type, such as Free or Pro.
- Expiry Date: The date the subscription expires.

7. Usage Log

This records when users access certain features, useful for analytics and billing.

It shows:

- Log ID: A unique ID for each log entry.
- User ID: The user who used the feature.
- Feature: The feature that was used.
- Timestamp: When it was used.

5.2 Use Case Diagram

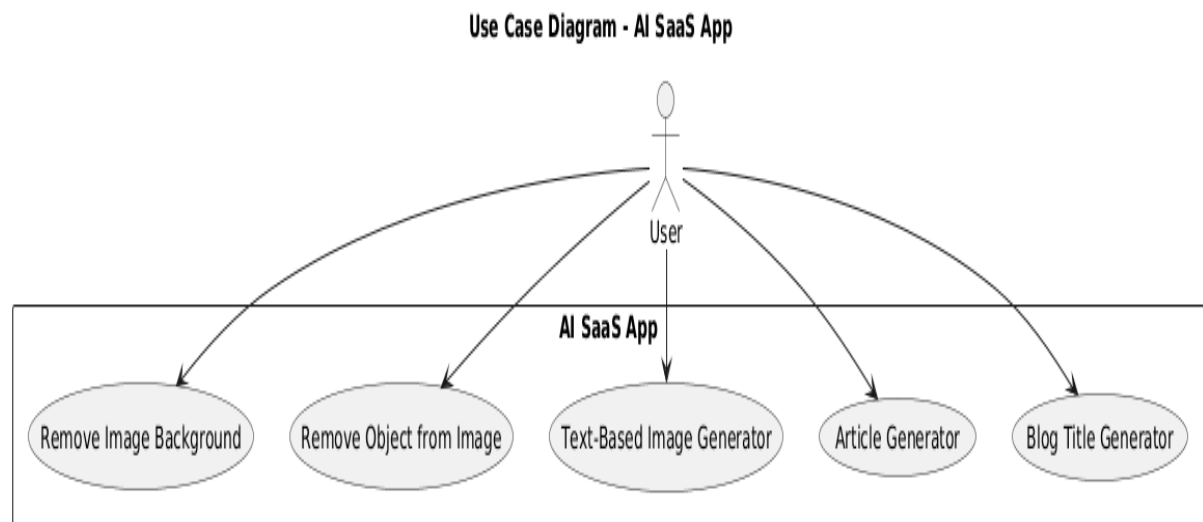


Figure 5.2

Purpose

This use case diagram shows what the AI SaaS application does from the user's view. It points out the main ways the user and the system work together, grouped by what the app can do.

Who's Involved: User

This is any signed-up user who uses the app through the website. They start all the actions. System Boundary:

What the System Does:

All the actions happen inside the AI SaaS application. The system takes care of:

- Checking what's entered.
- Sending things to the right parts of the app.
- Getting the results ready and sending them back.

5.3 Entity-Relationship Diagram

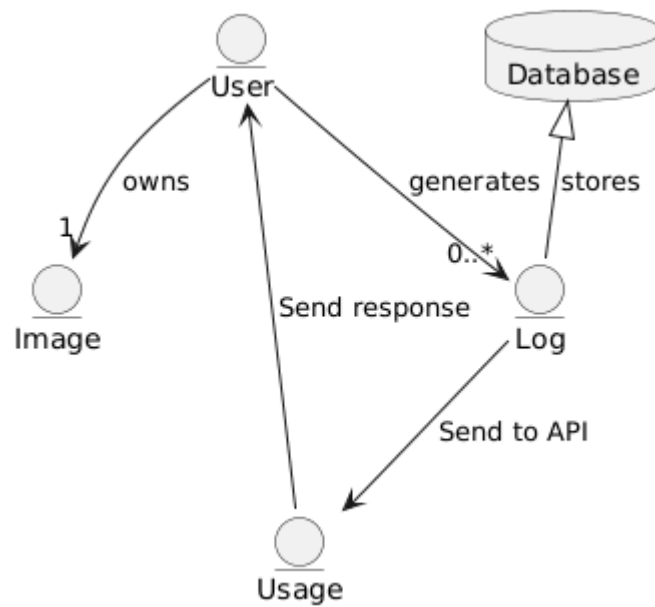


Figure 5.3

5.3.1 Goal

This ERD shows how our AI SaaS app's database is set up.

5.4 Sequence Diagram

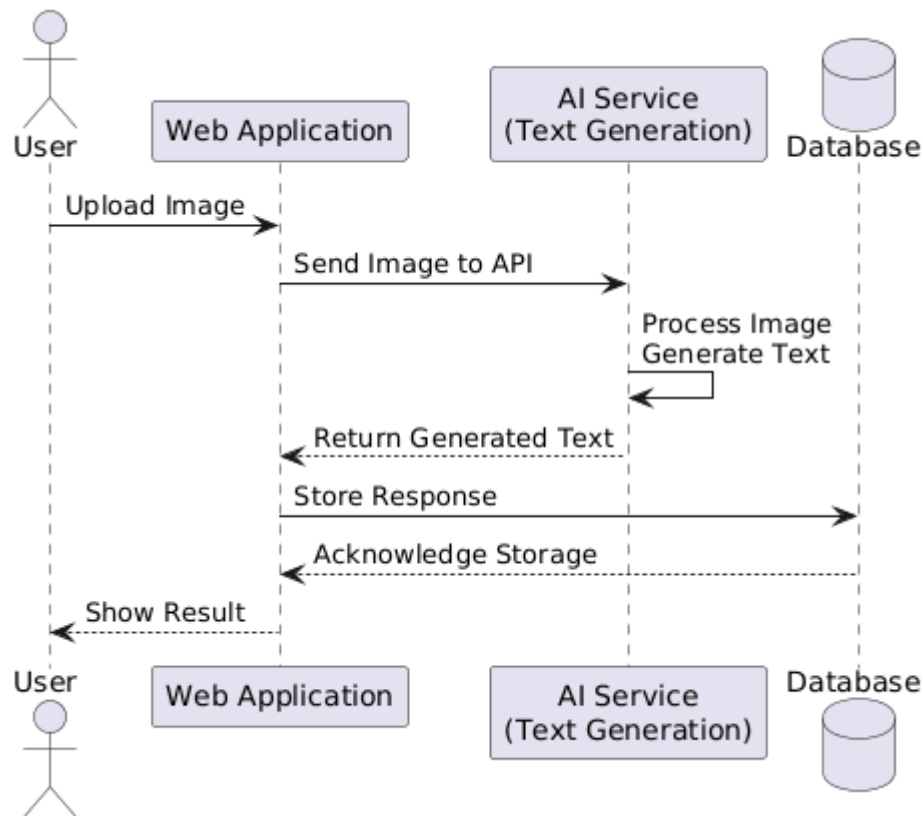


Figure 5.4

5.4.1 Purpose

This sequence diagram shows how the different parts of the system talk to each other when someone does something, like upload a photo or ask for some text to be written.

AI Service:

Does the main AI work?

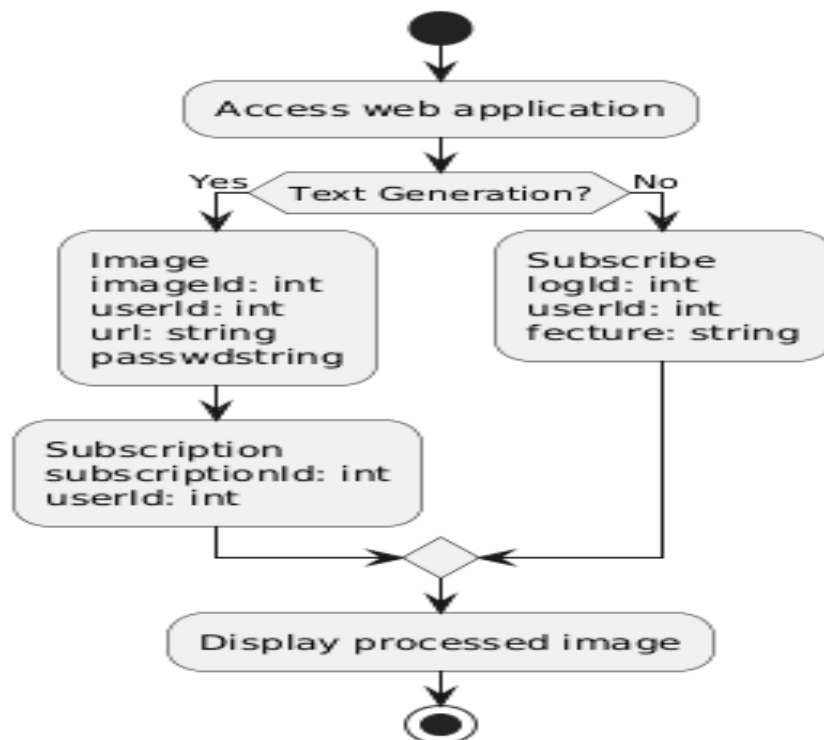
- Takes out the background of an image.
- Removes objects from an image.
- Turns text into an image.
- Creates titles for articles or blogs.
- Sends the finished product back to the API server.

Database:

- Keeps track of details, what users have done, and logs.
- Can be checked to find old results or see how much the system is being used.

How it works step by step:

1. User → Web Application: Uploads a picture or types in a question.
2. Web Application → API Server: Sends the request with the data.
3. API Server → AI Service: Sends the request to be processed.
4. AI Service → API Server: Sends the finished result back.
5. API Server → Web Application: Sends the response to the website.
6. Web Application → User: Shows the result (image or text).
7. API Server → Database: Logs the request and the details of the result.

5.5 Activity Diagram**Figure 5.5**

Purpose

This activity diagram shows how our AI software works.

6. IMPLEMENTATION

SCREENSHOTS

6.1 WELCOME SCREEN:

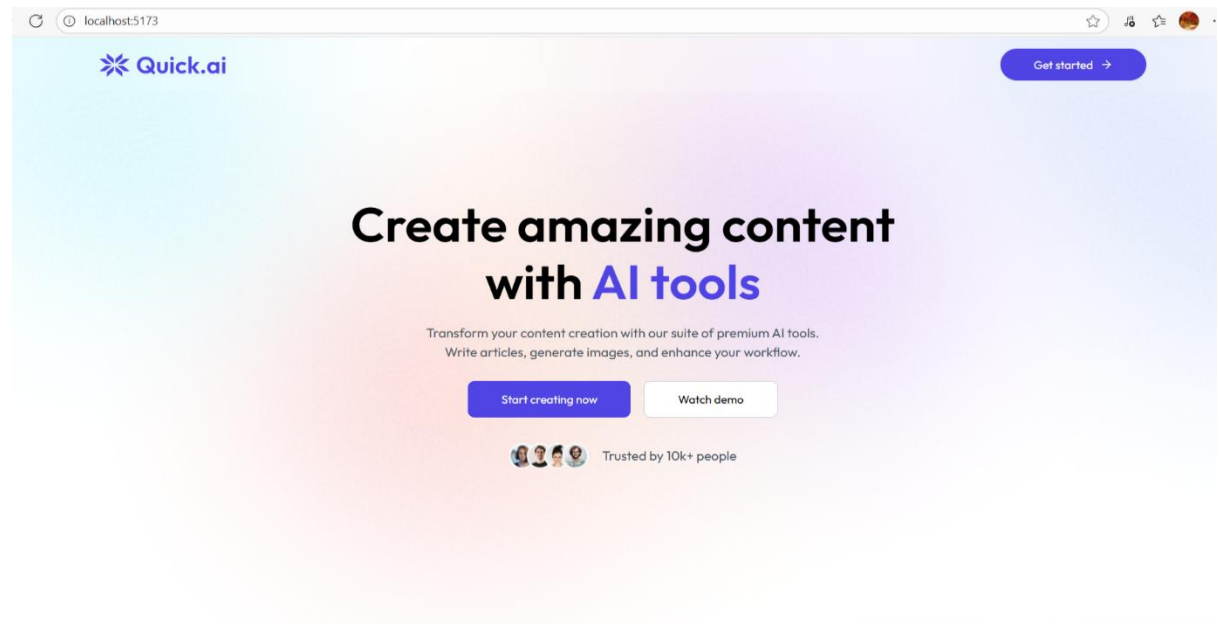


Figure 6.1

6.2 SIGN-IN SCREEN:

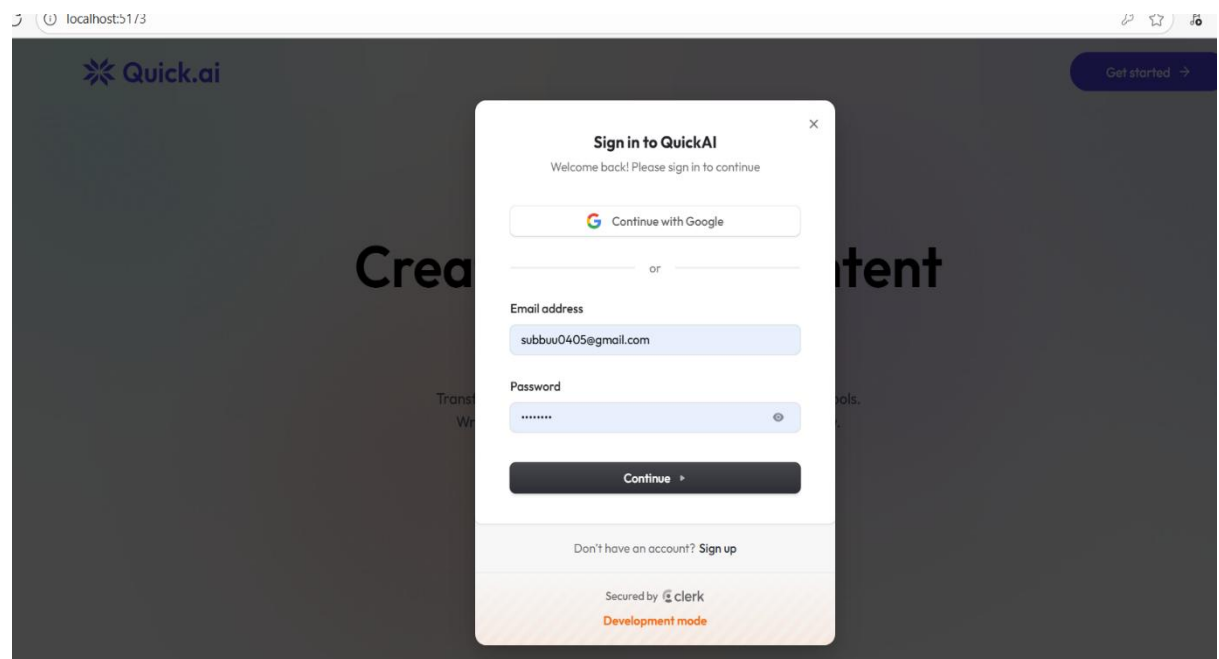


Figure 6.2

6.3 SIGN-UP SCREEN:

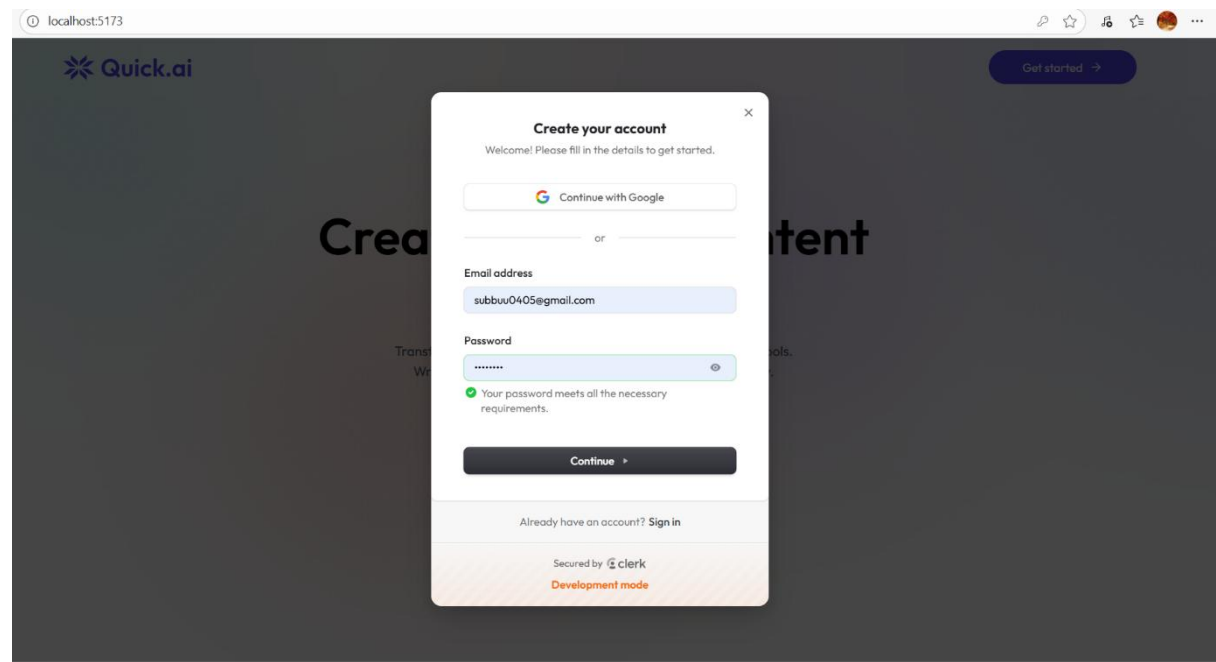


Figure 6.3

6.4 DASHBOARD:

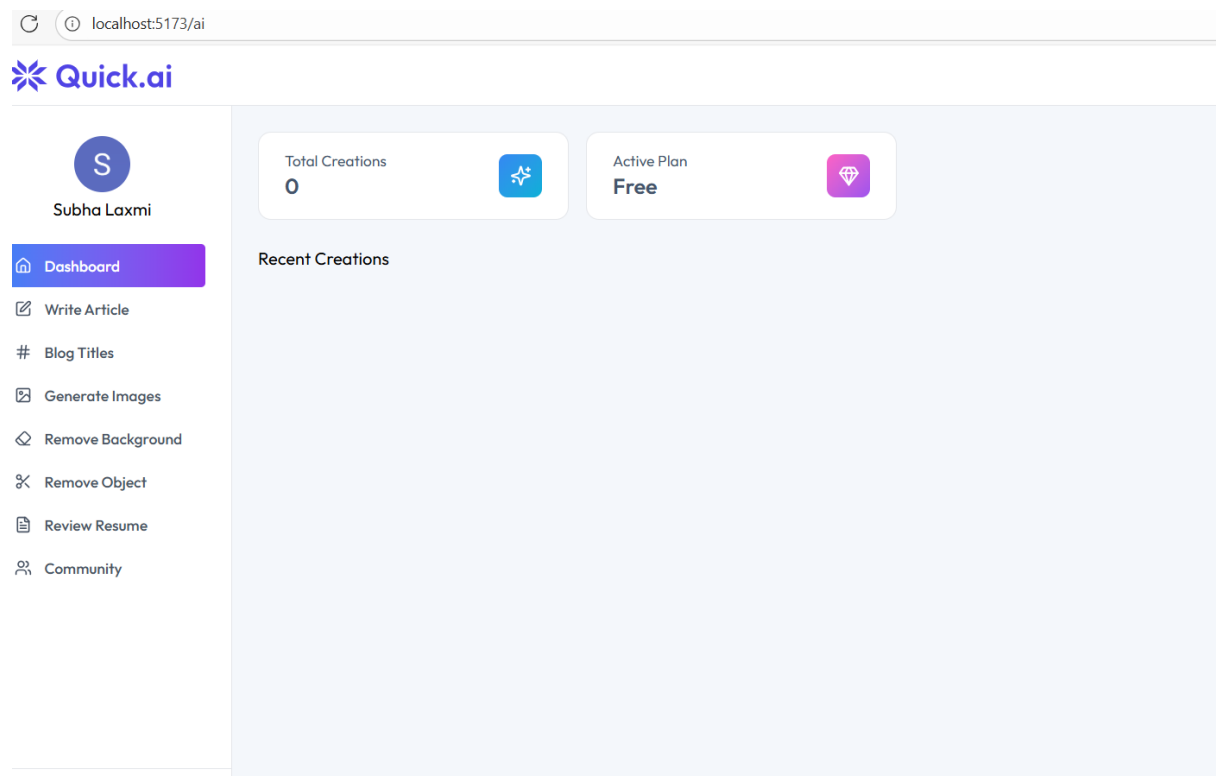


Figure 6.4

6.5 BLOG TITLE GENERATOR:

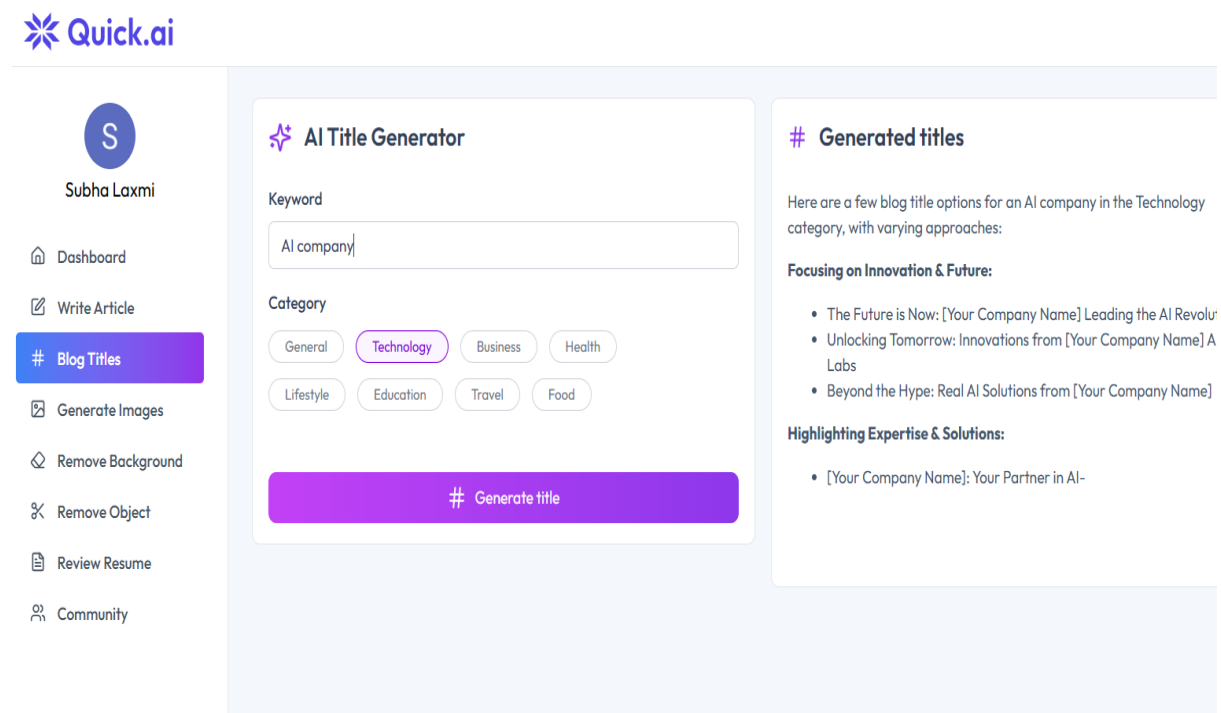


Figure 6.5

6.6 ARTICLE GENERATOR:

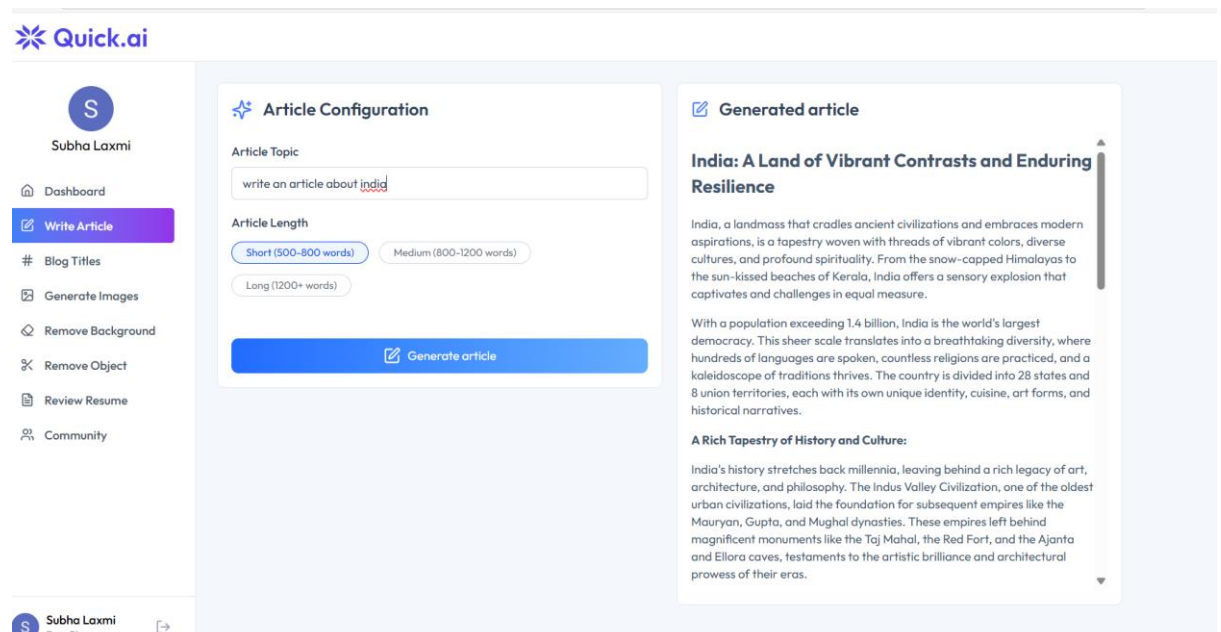


Figure 6.6

6.7 IMAGE GENERATOR:

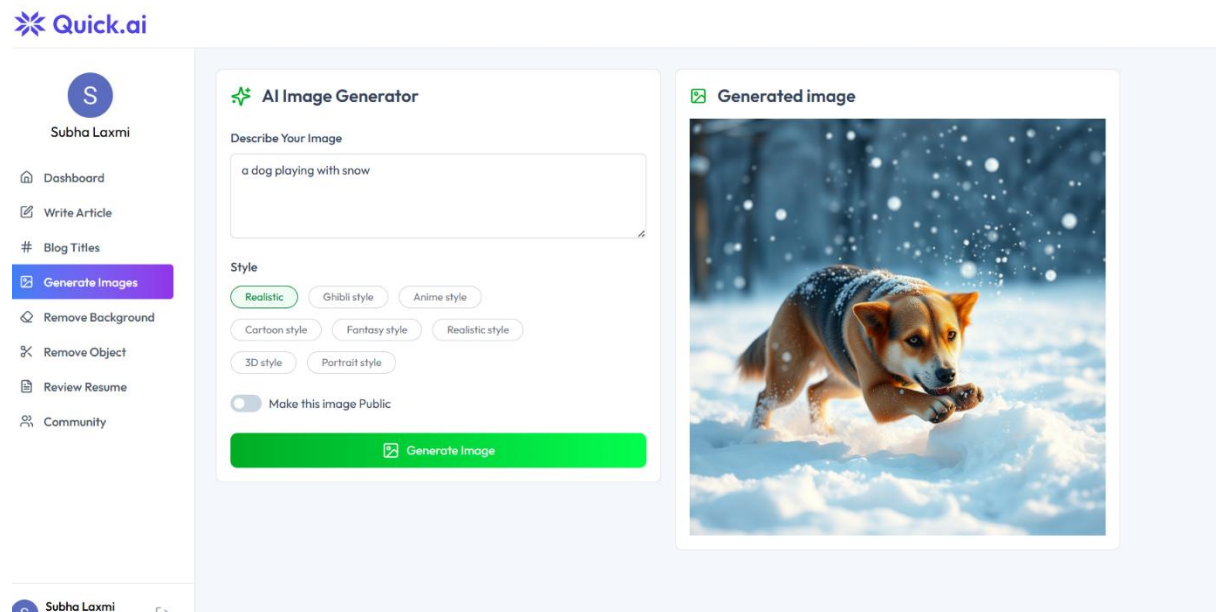


Figure 6.7

6.8 BACKGROUND REMOVER SCREEN:

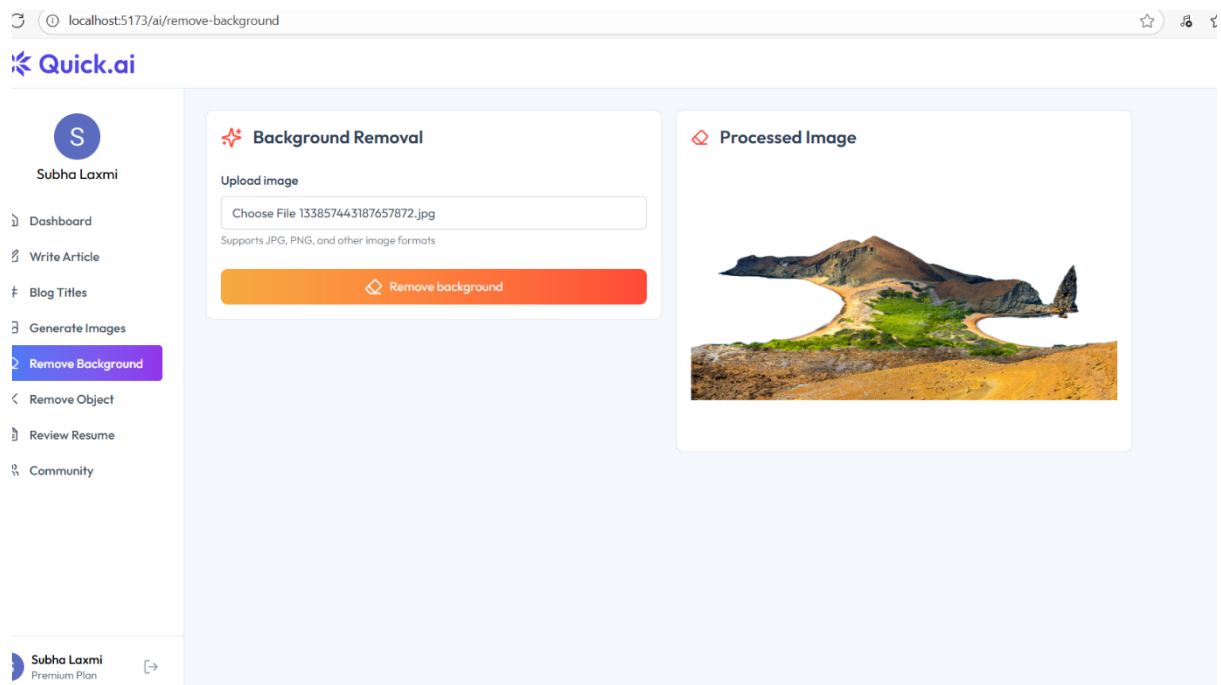


Figure 6.8

6.9 OBJECT REMOVER SCREEN:

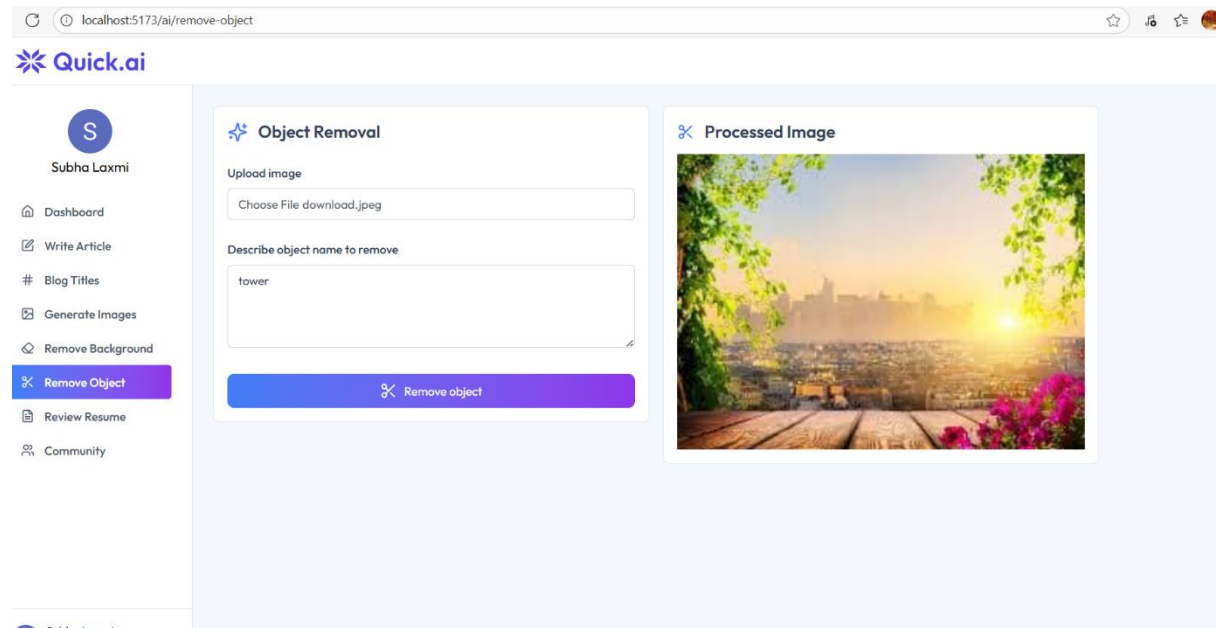


Figure 6.9

6.10 RESUME ANALYSIS SCREEN:

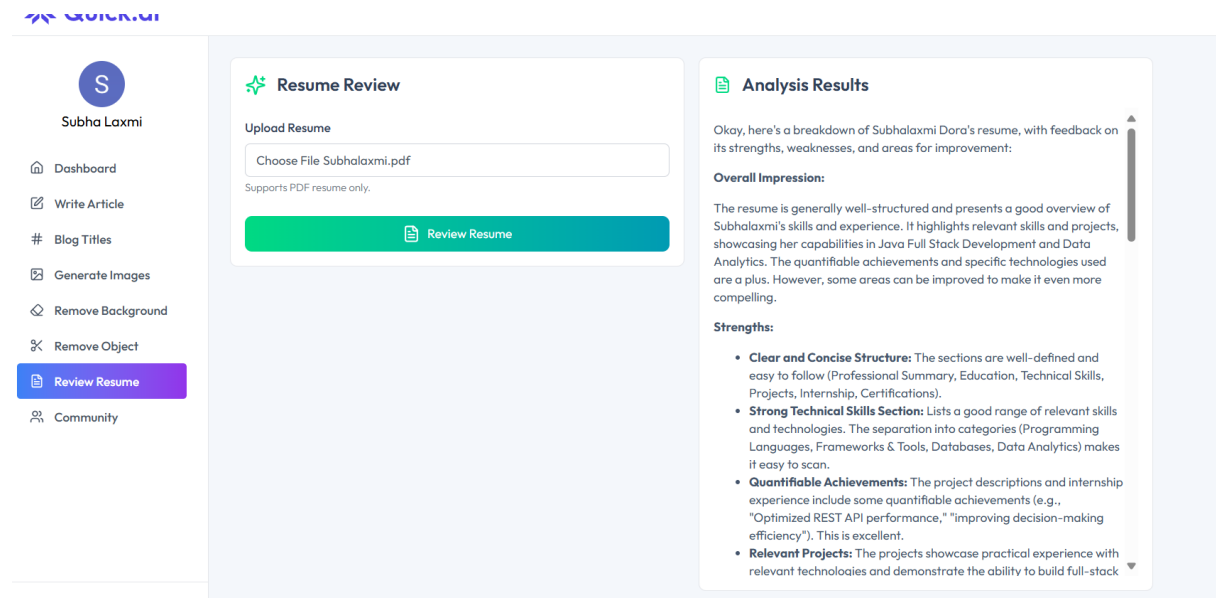
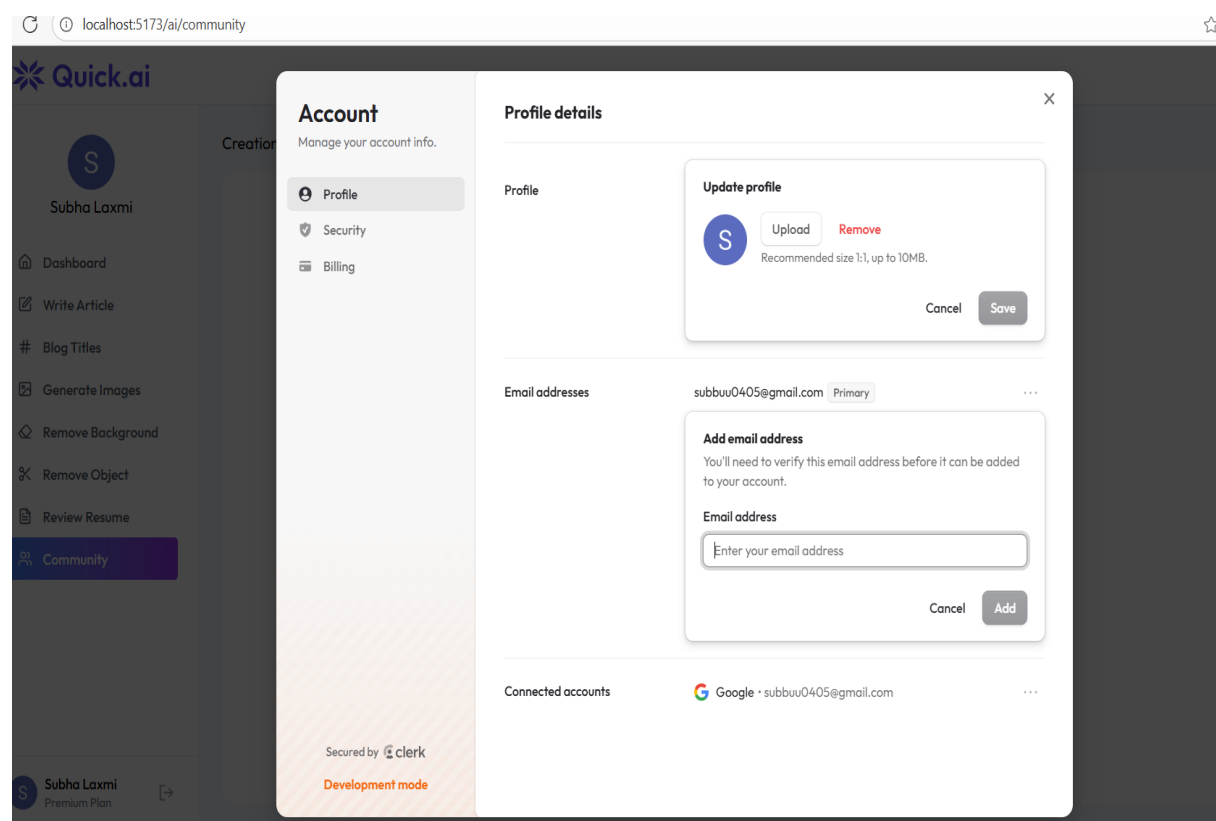
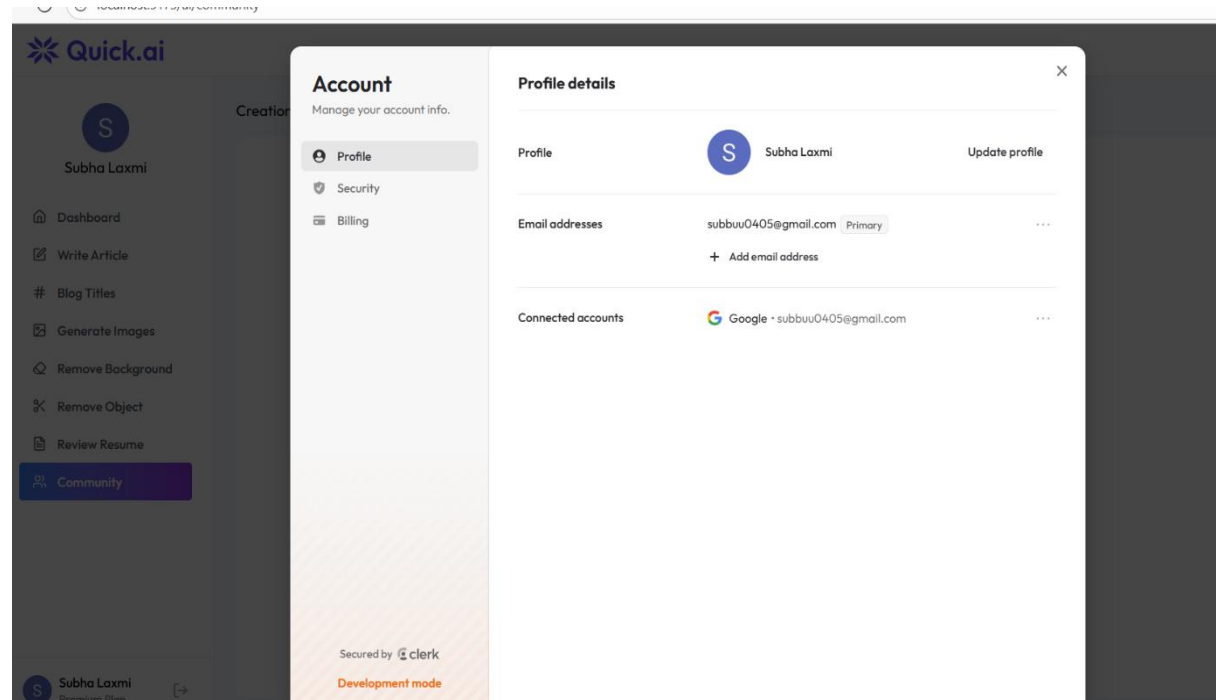
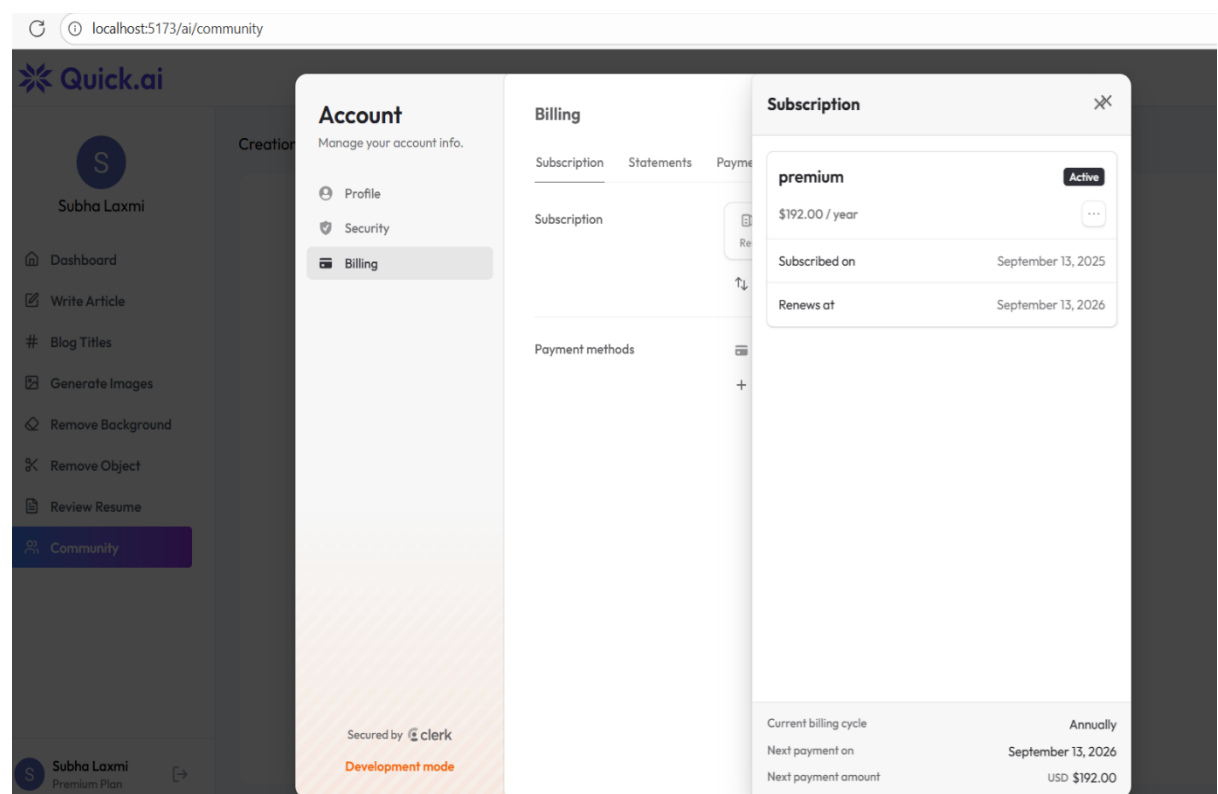
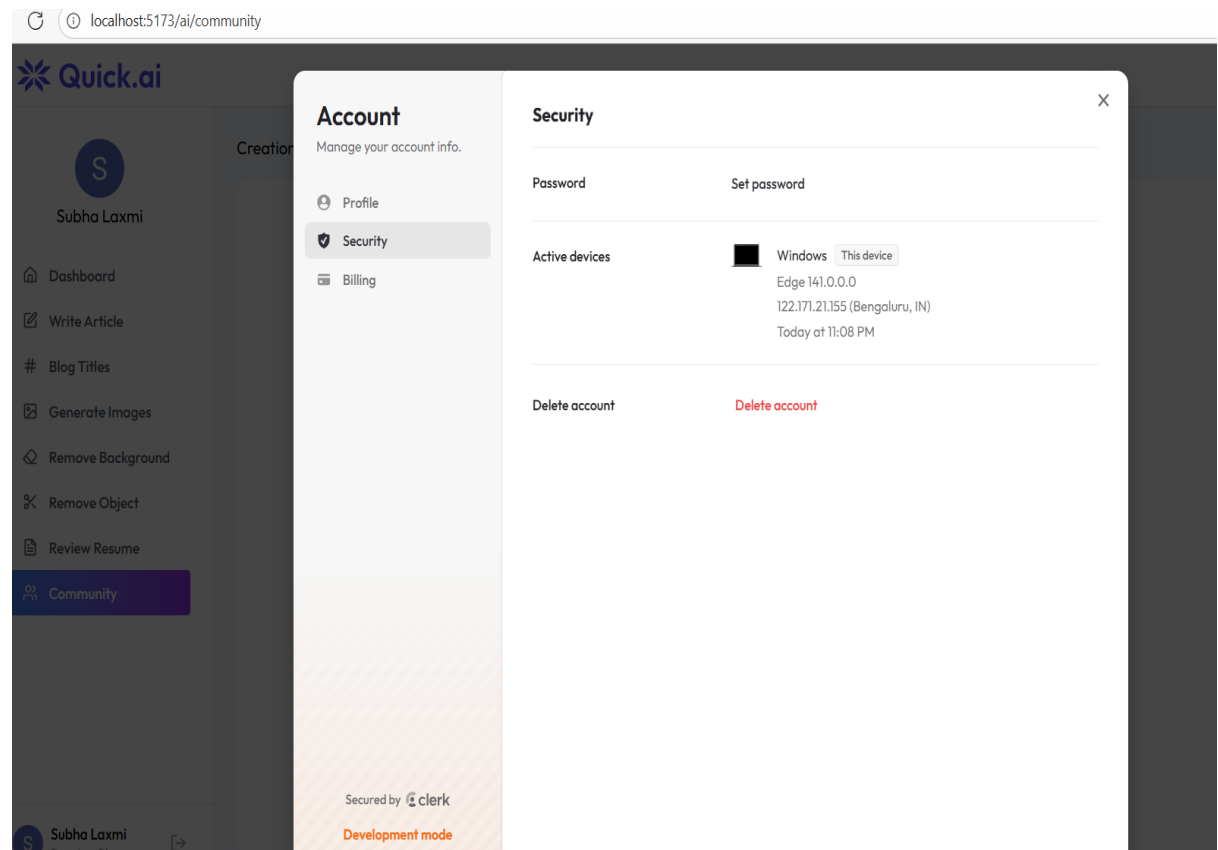


Figure 6.10







Powerful AI Tools

Everything you need to create, enhance, and optimize your content with cutting-edge AI technology.



AI Article Writer

Generate high-quality, engaging articles on any topic with our AI writing technology.



Blog Title Generator

Find the perfect, catchy title for your blog posts with our AI-powered generator.



AI Image Generation

Create stunning visuals with our AI image generation tool. Experience the power of AI.



Background Removal

Effortlessly remove backgrounds from your images with our AI-driven tool.



Object Removal

Remove unwanted objects from your images seamlessly with our AI object removal tool.



Resume Reviewer

Get your resume reviewed by AI to improve your chances of landing your dream job.



Choose Your Plan

Start for free and scale up as you grow. Find the perfect plan for your content creation needs.

Free

\$0

Always free

- ✓ Article Generation
- ✓ Generate Images
- ✓ Remove Background

Switch to this plan

premium

Active

\$16

/ month

☒ Billed annually

- ✓ Title Generation
- ✓ Article Generation
- ✓ Generate Images
- ✓ Remove Background
- ✓ Remove Object
- ✓ Resume Review

Project Code

Front end: React.js

App

```
import React from 'react'

import {Route, Routes} from 'react-router-dom'

import Home from '/pages/Home'

import Layout from '/pages/Layout'

import Dashboard from '/pages/Dashboard'

import WriteArticle from '/pages/Write Article'

import BlogTitles from '/pages/BlogTitles'

import GenerateImages from '/pages/GenerateImages'

import RemoveBackground from '/pages/RemoveBackground'

import RemoveObject from '/pages/RemoveObject'

import ReviewResume from '/pages/ReviewResume'

import Community from '/pages/Community'

import {useAuth} from '@clerk/clerk-react'

import {useEffect} from 'react'

import {Toaster} from 'react-hot-toast'

const App = () => {

  return (

    <div>

      <Toaster />

      <Routes>

        <Route path="/" element {<Home />}/>
```

```
<Route path='/ai' element {<Layout />}>

<Route index element={<Dashboard/>}/>

<Route path='write-article' element {<WriteArticle/>}/>

<Route path='blog-titles' element {<BlogTitles/>}/>

<Route path='generate-images' element {<GenerateImages/>}/>

<Route path='remove-background' element {<RemoveBackground/>}/>

<Route path='remove-object' element {<RemoveObject/>}/>

<Route path='review-resume' element {<ReviewResume/>}/>

<Route path='community' element {<Community/>}/>

</Route>

</Routes>

</div>

)
```

```
export default App
```

Main

```
import {createRoot} from 'react-dom/client'

import './index.css'

import App from './App.jsx'

import {BrowserRouter} from 'react-router-dom'

import {ClerkProvider} from '@clerk/clerk-react'

// Import your Publishable Key

const PUBLISHABLE_KEY = import.meta.env.VITE_CLERK_PUBLISHABLE_KEY
```

```
if (!PUBLISHABLE_KEY) {  
  
  throw new Error('Missing Publishable Key')  
  
}  
  
createRoot(document.getElementById('root')).render(  
  
  <ClerkProvider publishableKey={PUBLISHABLE_KEY} afterSignOutUrl="/">  
  
    <BrowserRouter>  
  
      <App />  
  
    </BrowserRouter>,  
  
    </ClerkProvider>  
  
  )  
)
```

Back-end:

aiRoutes.js

```
import express from "express";  
  
import {auth} from "/middlewares/auth.js";  
  
import {generateArticle, generateBlogTitle, generateImage, removeImageBackground,  
removeImageObject, resumeReview} from "../controllers/aiController.js";  
  
import {upload} from "../configs/multer.js";  
  
const ai Router = express.Router();  
  
aiRouter. Post ('/generate-article', auth, generateArticle)  
  
aiRouter. Post ('/generate-blog-title', auth, generateBlogTitle)  
  
aiRouter post ('/generate-image', auth, generateImage)  
  
aiRouterpost ('/remove-image-background', uploadsingl('image'),auth, removeImage  
Background)
```

```
aiRouterpost ('/remove-image-object', uploadsingle('image'), auth, removeImageObject)
```

```
aiRouterpost ('/resume-review', uploadsingle('resume'), auth, resumeReview)
```

```
export default aiRouter
```

Database: PostgreSQL

UserController.js

```
import sql from "../configs/db.js";
```

```
export const getUserCreations = async (req, res){
```

```
  try {
```

```
    const {UserId} = req.auth()
```

```
    const creations = await sql` SELECT * FROM creations WHERE user_id = ${user-Id}
    ORDER BY created at DESC`;
```

```
    res.json({success: true, creations});
```

```
  } catch (error) {
```

```
    res. Json ( {success: false, message: error message});
```

```
  }
```

```
}
```

```
export const get PublishedCreations = async(req, res){
```

```
  try {
```

```
    const creations = await sql
```

```
    SELECT * FROM creations WHERE publish = true ORDER BY created at DESC`;
```

```
    res.json({success: true, creations});
```

```
  } catch (error) {
```

```
    res.json({success: false, message: error message});
```

```
  }
```

```
}

export const toggle LikeCreation = async (req, res){

  try {

    const {userId} = req. auth ()

    const {id} = req body

    const [creation] = await sql SELECT * FROM creations WHERE id = ${id}`

    if(creation){

      return res.json({success: false, message: "Creation not found"})

    }

    const currentLikes = creation likes;

    const userId Str = user Id.ToString ();

    let updatedLikes;

    let message;

    if (currentLikes. Includes(userIdStr)) {

      updatedLikes = currentLikes.filter((user)=>user userIdStr);

      message = 'Creation Unliked'

    } else {

      UpdatedLikes = [...currentLikes, userIdStr]

      message = 'Creation Liked'

    }

    const formattedArray = ${updatedLikes join (',')}

    await sql UPDATE creations SET likes = ${formatted Array}:text [] WHERE id =

    ${id};res.json({success: true, message});

  } catch (error) {

    res.json({success: false, message: error message});

  }

}
```

}

}

7. SOFTWARE TESTING

7.0 TESTING METHODS

7.0.1 UNIT TESTING:

The first step in software testing for the Quick AI SaaS app is unit testing. We test every component independently, such as the Background Remover Image Generator and Resume Analyzer, to ensure that the fundamentals function as intended.

For instance, we verify that the Resume Analyzer provides accurate ATS scores and formatting guidelines and that the Background Remover can truly distinguish between an object and a background.

7.0.2 INTEGRATION TESTING:

Integration testing is a crucial step in the development of the Quick AI SaaS app. This makes sure that the app's various components are all functioning properly together. Think about it: while each piece may be fine by itself, they also need to be able to communicate with one another without any issues. Suppose that a picture is uploaded with the background removed. the React-built portion of the application that they view. The request must be properly sent by JS to the Node.JS component that performs the work silently. JS/Express. After that, that section must communicate with the AI and retrieve the PostgreSQL database results.

Integration testing is about making sure different APIs and services work together correctly. Basically, it checks if the data they send and receive is compatible.

You can mimic API calls with tools like Postman or automated tests to see if the responses are accurate. Doing this helps you spot problems in how things connect, like data format issues, mismatches, or bad dependencies.

For example, integration tests make sure features in the Quick AI app, like the Blog Title Generator and the Article Generator, not only work on their own but also play nicely with everything else.

7.0.3 SYSTEM TESTING:

System testing makes sure the Quick AI SaaS app works right from start to finish. It checks if the app does what it's supposed to do in a real-world way. Unlike unit and integration tests that look at pieces of the app, system testing checks the whole thing. So, if someone logs in, uploads their resume, gets it checked, writes an article, and saves it, we test if that whole process works.

This makes sure everything functions and is quick dependable and user-friendly. Computers or people can be used to replicate user actions and verify that the results are correct. System testing for Quick AI ensures that storing data putting it on the cloud and verifying who is using it all function without any problems. It identifies major issues like errors in the way things flow when things do not work together or when we forget something. This indicates that users can begin using the app.

7.0.4 USER ACCEPTANCE TESTING (UAT):

UAT ensures that the Quick AI SaaS App performs as users would expect and desire. Following technical testing to ensure everything functions as intended UAT assesses the products usability and suitability for business requirements. Here actual users test the app in authentic scenarios such as uploading resumes creating blog titles or extracting objects from images.

According to their opinions the system is easy to use fast and produces accurate results. For example; a recruiter should find the Resume Analyzer helpful in gathering information. Good SEO-friendly titles should be generated for bloggers who use the Blog Title Generator. UAT identifies any discrepancies between user satisfaction and the apps functionality. This ensures that the app is useful in addition to functioning. Prior to releasing the app this step is crucial. It demonstrates that the platform is user-friendly and prepared for the market.

7.0.5. PERFORMANCE TESTING:

We run performance tests to ensure the Quick AI SaaS app can handle large numbers of users at once without stuttering or crashing. Many people will use it collectively because it is on the cloud so we must make sure it is stable fast and quick.

The system should still function in a matter of seconds even if 100 users attempt to create articles or remove backgrounds simultaneously. Load testing demonstrates the behaviour

of the application under typical and extremely busy conditions. We can determine when it breaks by stress testing.

Scalability testing makes sure we can add more servers or stuff to handle more people when the app grows. The apps performance can be observed by simulating thousands of users using tools such as JMeter or Locust. Through these tests Quick AIs developers can ensure that the system is dependable fast and always up even when a large number of users are utilizing it. In this manner everyone enjoys using the app which is crucial for any SaaS platform.

7.0.6 SECURITY TESTING:

Because our platform handles private stuff like resumes and personal info, we have to follow data rules like GDPR.

Verifying that our login process functions properly and that only legitimate users are granted access is known as security testing. To ensure that all information sent between the app its brain and the database remains, secure we also examine how we encrypt data. We employ scans and pen testing to identify vulnerabilities that hackers might exploit.

For instance, we attempt to sneak into files we should not inject malicious scripts or circumvent the login. By avoiding leaks or corrupted files security testing also ensures that our data storage in PostgreSQL is secure.

The Quick AI app is a trustworthy safe and legally sound platform because of its robust security which also helps it defend against online threats and gain user trust.

7.1 TEST CASES:

Test Case ID	Module/Feature	Test Scenario	Input	Expected Output	Type	Status
TC-01	Background Remover	Upload an image and remove the background .	Image with an object and background.	Background gone, object easy to see.	Functional / Unit	Pass
TC-02	Background Remover	Upload a file that's not supported	.txt file	Error message saying "Invalid file format".	Negative Testing	Pass
TC-03	Image Generator	Make an image from text	"A dog playing football"	AI-creates an image that matches what the text says	Functional / System	Pass
TC-04	Image Generator	Enter nothing for the prompt.	Blank text	Error message "Prompt cannot be empty".	Negative Testing	Pass
TC-05	Object Remover	Get rid of something you do not	Photo with watermark/logo.	Item is removed, and	Functional / Unit	Pass

		want in an image.		the spot looks natural.		
TC-06	Object Remover	Try to remove an object from a bad image.	Blurry image	Part of it is removed.	Functional / Edge	Pass
TC-07	Resume Analyzer	Look at an uploaded resume.	PDF resume	Shows ATS score, keyword matches, tips on formatting	Functional / UAT	Pass
TC-08	Resume Analyzer	Upload a broken file.	Corrupted resume file.	Error message "File not supported or corrupted"	Negative Testing	Pass
TC-09	Article Generator	Make an article from a topic	"AI in Healthcare"	Easy-to-read article with clear headings	Functional / System	Pass
TC-10	Article Generator	Try to generate an article with something weird.	"@@@@@"	Error message "Invalid input"	Negative Testing	Pass
TC-11	Blog Title Generator	Suggest blog titles.	Blog topic: "Cloud Security Trends".	Gives 5–10 SEO-friendly, creative title ideas	Functional / Unit	Pass

TC-12	Blog Title Generator	Enter nothing as the topic	Blank input	Error message “Topic cannot be empty”	Negative Testing	Pass
TC-13	Authentication	user logs in with the right info.	Correct email and password.	Logs in and goes to the dashboard.	Security / Functional	Pass
TC-14	Authentication	User logs in with the wrong info.	Wrong password	Error message “Invalid login information”	Security / Negative	Pass
TC-15	System Performance	What happens when 100 users use it at once?	100 requests At the same time	System answers in less than 3 seconds and doesn't crash	Performance Testing	Pass
TC-16	Database Security	Save Data.	Resume/ image upload	Data is saved safely in PostgreSQL with encryption.	Security / Integration	Pass
TC-17	API Integration	Check the API response for the article generator.	REST API call with a topic	JSON response with the generated article.	Integration Testing	Pass

TC-18	Usability	Does it work on mobile?	Open the app on a phone browser.	The interface looks right. Nothing is overlapping.	Usability Testing	Pass
TC-19	Error Handling	What happens if the system crashes?	Force a backend error	The user sees “Server error: please try again later”.	Reliability Testing	Pass
TC-20	Accessibility	Does it work for users with disabilities?	The Screen reader is on.	The app is readable and easy to use with accessibility tools.	Accessibility Testing	Pass

Table 7.2

8. CONCLUSION

Quick AI SaaS Application is a big move in using artificial intelligence to fix problems we see every day. The project puts together six cool tools—background remover, image maker, object remover, resume checker, article maker, and blog title maker—all in one place online. These tools help all sorts of people, from making creative stuff to building good resumes, showing that AI can be handy for everyday things.

We worked hard to make Quick AI easy to use, fast, and able to grow. The way it is built makes sure it runs well, and the simple design means anyone can use it. Plus, since it is a SaaS, you can get to it easily and do not need a ton of equipment to run it.

This project shows that AI can really help make us more productive, cut down on boring work, and boost our imagination. Right now, it does what we wanted it to, but there's still room to make it even better. We could add a phone app, support more languages, connect it to other programs, and make it more personal.

Basically, Quick AI proves that you can take artificial intelligence and turn it into a helpful SaaS platform for everyone—students, individuals, and businesses. If we keep working on it, it could become a full-blown AI system that really helps move us into the digital age.

9. FUTURE ENHANCEMENTS

To keep Quick AI a useful and competitive tool, we need to keep making it better. Even though the current version of our SaaS app has cool stuff like a background remover, AI image creator, object remover, resume analyser, article creator, and blog title generator, there's still room to grow. By improving Quick AI, we can make it easier to use, more reliable, and able to handle whatever comes its way in the long run.

Here are some ideas:

9.1 Add More AI Tools

Right now, Quick AI has six smart features. We could add more, like voice-to-text, text summarization, AI chatbots, plagiarism checkers, and automatic video creation. This would turn Quick AI into a complete AI toolkit for students, professionals, and businesses.

9.2 Create a Mobile App

Quick AI is currently a web-based app. To make it easier to get to, we could make a mobile app for Android and iOS. People could then use the AI tools on their phones or tablets, anytime, anywhere, which would get more people using the platform.

9.3 Make it More Personal

Future versions could use machine-learning to learn what users like. If someone uses Quick AI to make resumes or articles a lot, the system could suggest templates, writing styles, and ways to improve their work. This would make the user experience way better by giving people exactly what they need.

9.4 Improve the Look and Feel

We could create a more user-friendly and attractive design to make Quick AI easier to use. Imagine drag-and-drop features, real-time previews, a dark mode, and dashboards that you can change to fit your needs. A modern look and feel would draw in both beginners and experts.

9.5 Make it Scale and Streamline the Cloud

As more people use Quick AI, it is really important that it remains available and scalable. To do it, we can consider using serverless setups, containerization, and fine-tuned load

balancing. So that Quick AI runs well even when lots of people are using it and can grow without problems.

9.6 Add Ways to Collaborate

What if Quick AI allowed multiple users to work on a project at the same time? For example, a team of writers could work on an article together, or recruiters could analyze resumes as a group. These features would make Quick AI good for professional and business use.

9.7 Boost Data Security and Privacy

Since Quick AI handles personal data like resumes and user-created content, security is super important. We could add end-to-end encryption, control who has access to what, follow GDPR rules, and use AI to spot threats. This would help users trust us and keep their info safe.

9.8 Offer Multiple Languages

Right now, Quick AI might only be in English. To reach more people, we could add support for languages like Hindi, Spanish, French, Chinese, and Arabic. This would allow people all over the world to use Quick AI, not just English speakers.

9.9 Connect to Other Platforms

We could make Quick AI work with other platforms like LinkedIn, WordPress, Blogger, and social media. For example, a blog title or article created by Quick AI could be posted directly to WordPress. Also, resumes could be sent straight to LinkedIn profiles. This would save time and make things easier for users.

10. Track Performance Automatically

Lastly, we can improve Quick AI with AI-driven dashboards to keep tabs on how things are going. Users and admins could see how the system is being used, how well the outputs are generated, and feedback reports. We can use this data to constantly improve the algorithms and improve the experience for our users.

Appendix - A

BIBLIOGRAPHY

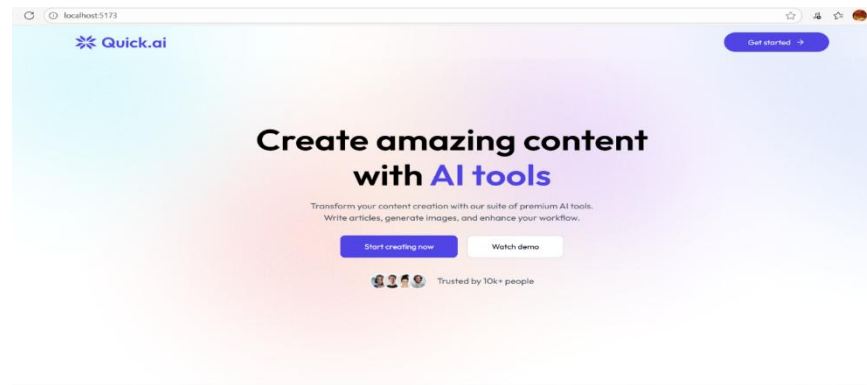
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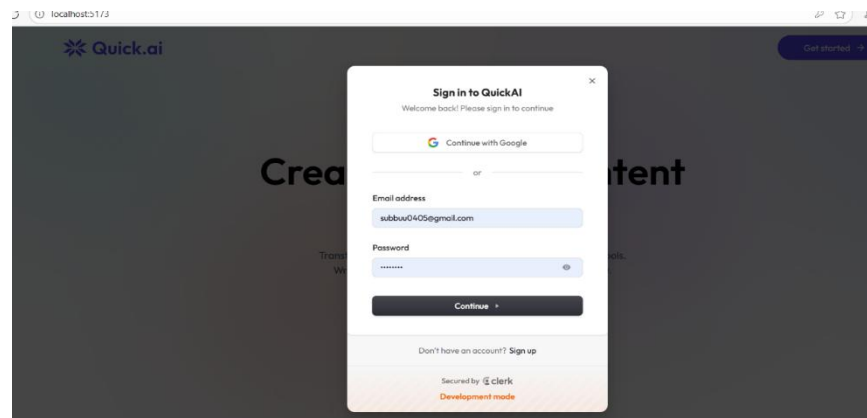
Appendix – B

USER MANUAL

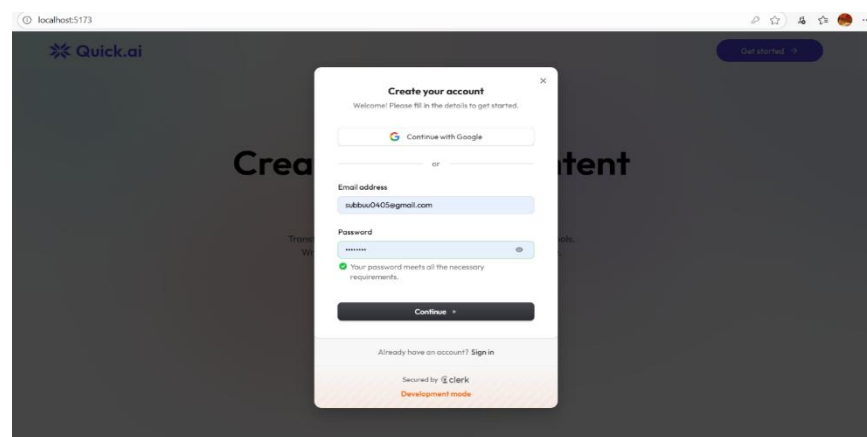
1 WELCOME SCREEN:



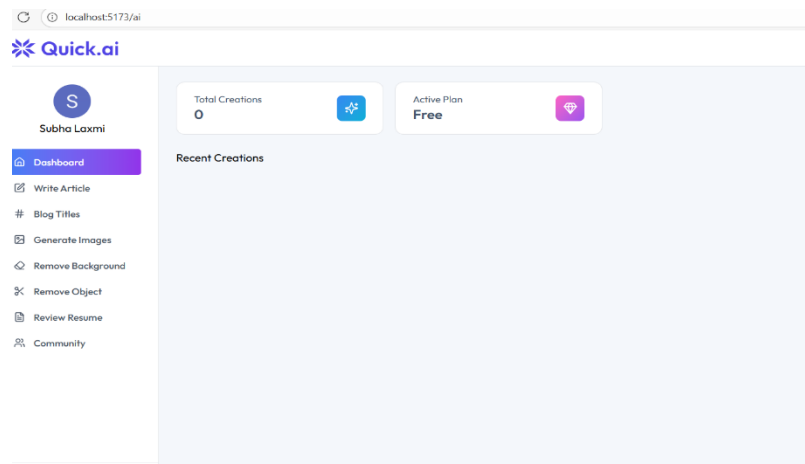
2 SIGN-IN SCREEN:



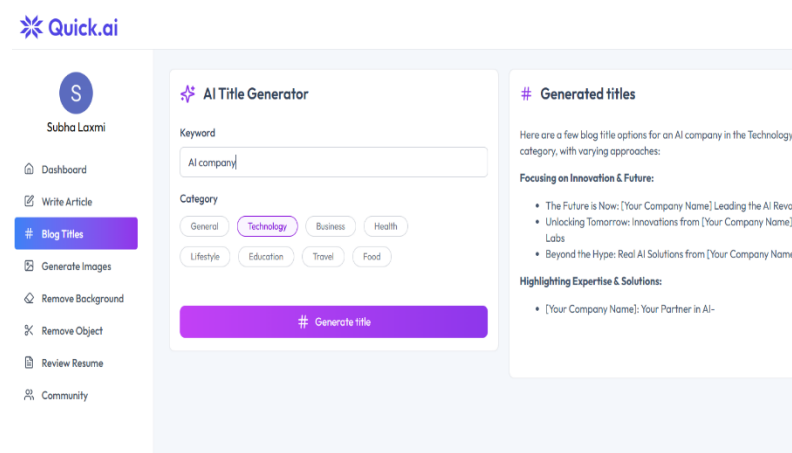
3 SIGN-UP SCREEN:



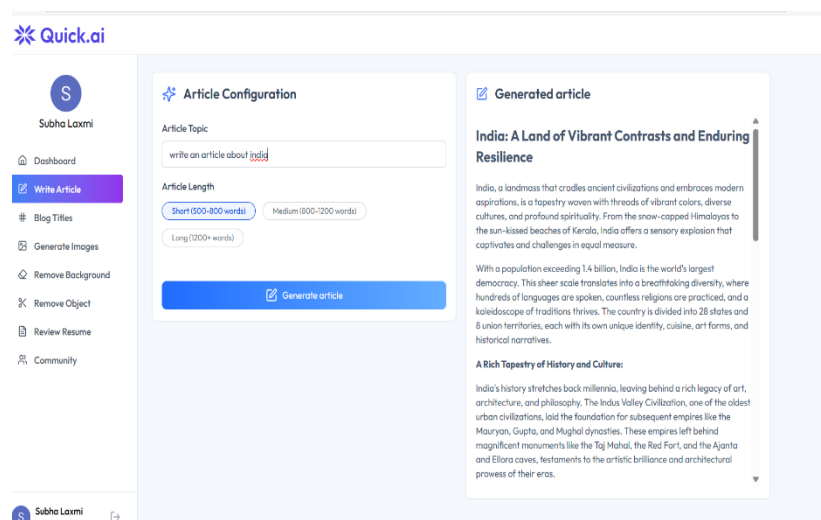
4 DASHBOARD:



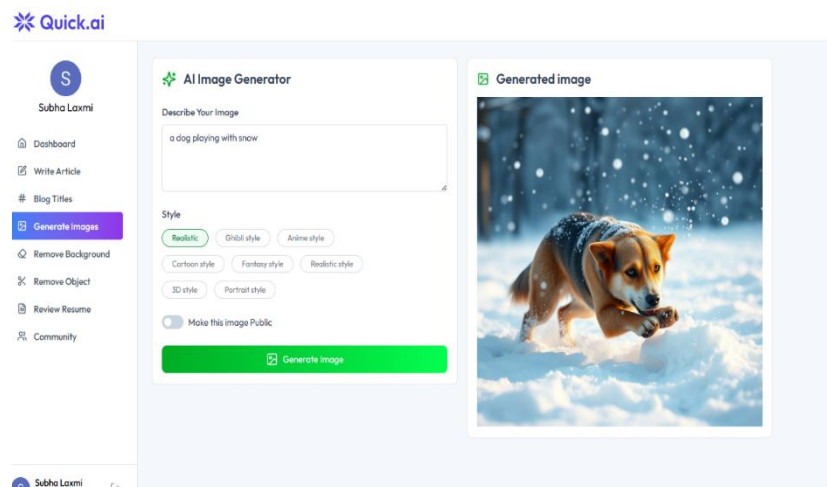
5 BLOG TITLE GENERATOR:



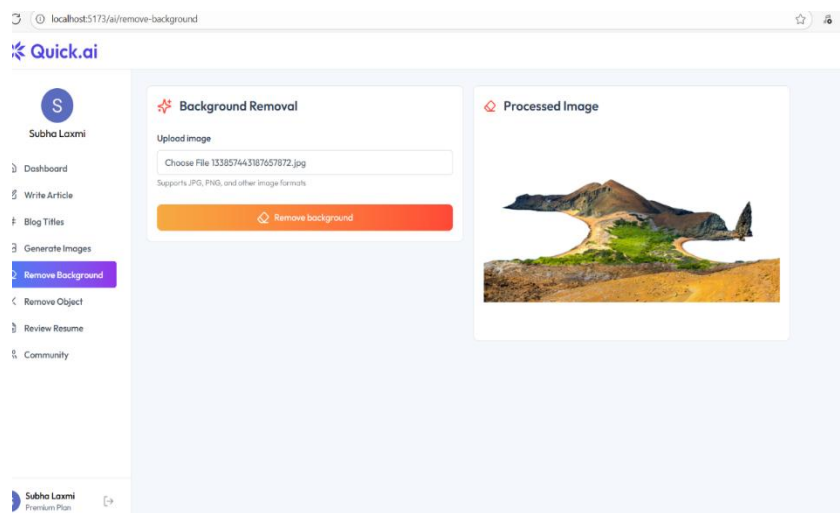
6 ARTICLE GENERATOR:



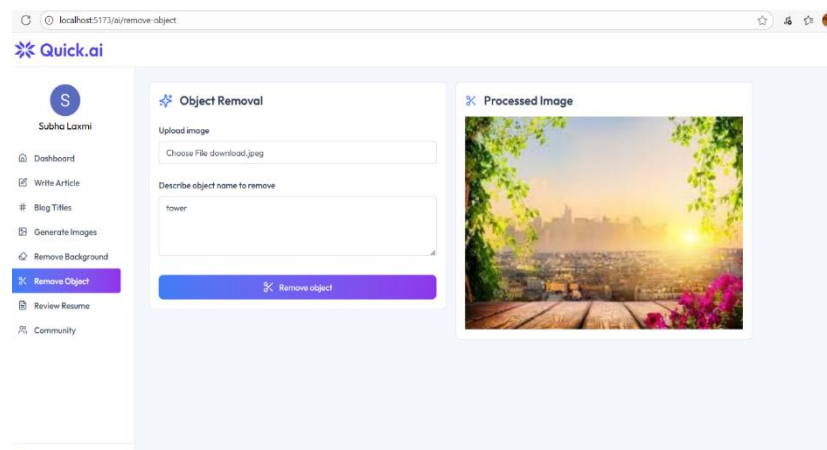
7 IMAGE GENERATOR:



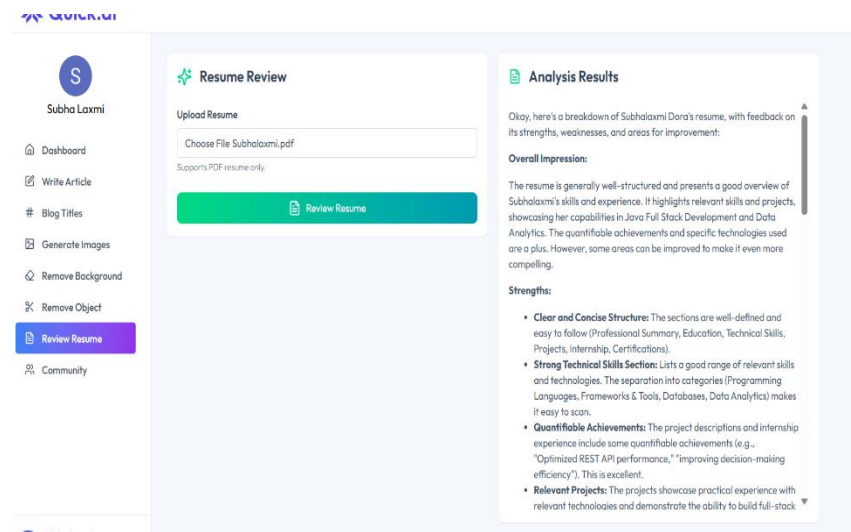
8 BACKGROUND REMOVER SCREEN:



9 OBJECT REMOVER SCREEN:



10 RESUME ANALYSIS SCREEN:



1 WELCOME SCREEN:

This screen provides the user interface for the SaaS App and gives the testimonial that has given by the user. It gives the button to start sign-in, sign-up and provides the information about the website.

2 SIGN-IN SCREEN:

In this screen we can see it is asking about the user email and password. It makes the successful Signing in of the user where they can use the services that is provided by the site. The continue button will take the user to next page.

3 SIGN-UP SCREENS:

This screen will create an account of the user which will then be stored in the database. After successfully creating the account the user can get the service and make the use of it.

4 DASHBOARDS:

In this picture we can see the dashboard of the app where it shows the multiple features by showing the user information and the plans with credits. The features that are showing as article generator, blog generator, image generator, resume analysis etc.

5 BLOG TITLE GENERATOR:

In this blog title generator screen, we can see that if user provides the keyword, then AI will generate the blog title according to user requirement which have list of categories.

6 ARTICLE GENERATOR:

In this screen if the user provides the topic for the article, It will provide the different lengthy articles. Where user is also asked about the length that he wants to be generate.

7 IMAGE GENERATOR:

In this image as we can see in the describe your image box if user gives the input, then according to user requirement it will generate the image, also it has multiple categories for the image.

8 BACKGROUND REMOVER SCREEN:

In this, user could provide the image in upload image box and after pressing the button the image will be shown without any background.

9 OBJECT REMOVER SCREEN:

In this feature of quick AI app where the user needs to provide image in image box and needs to mention object name inside the other box. After pressing the remove background button user can get the image without the mentioned object.

10 RESUME ANALYSIS SCREEN:

In this screen we can see the user needs to provide the resume which he needs to analyse and then user needs to press the button review resume. In analysis screen it will show the strengths and weakness of the resume with a detailed review of the resume.